OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

Abstract. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Numerical estimates of the value and gradient are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity.

For nonhomogeneous activation functions on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$, we study the normalized objective

$$J(\mu) = L(\mu) + \alpha\Phi_\phi(\mu_p),$$

where Φ_ϕ is induced by a concave scalar penalty ϕ with finite positive right slope at zero, and $\mu_p = w_p\mu$ with $p > 0$ and $w_p(\omega) = 1 + |\omega|^p$. Under polynomial growth, choosing p sufficiently large makes the normalized dictionary vanish at infinity. This yields existence of the global minimum by normalized weak-* compactness. If the penalty is strongly concave on every bounded interval, every local minimizer has finite support and satisfies a covering-number bound. We propose a Primal-Dual-Active-Point (PDAP) algorithm. For the single-atom PDAP scheme with exact search and monotone correction, we prove an $O(n^{-1/2})$ best-iterate bound on the first-order optimality residual for atom insertion.

For positively k-homogeneous activations, we give an equivalent measure problem on the unit sphere with the fractional-exponent penalty $|c|^{2/(k+1)}$ on the outer coefficients. For ReLU^k with $k \geq 2$, every local minimizer has finite support, while every global minimizer satisfies an explicit a priori width bound. An insertion algorithm based on the optimality condition is proposed.

We study the resulting algorithms on the Van der Pol oscillator and pendulum swing-up benchmarks; the latter has a value function with a discontinuous gradient.

Contents

1. Introduction 2
2. Preliminaries 4
2.1. Finite signed Radon measures 5
2.2. Measures with a finite moment 5
Notation 5
3. The learning problem 5
3.1. The value operator and the gradient-augmented loss 5
3.2. The nonconvex functional 6
3.3. The normalized problem 11
3.4. Existence of a global minimizer 13
3.5. Optimality conditions and finite support 14
4. The k-homogeneous formulation 21
4.1. The measure problem on the sphere 21
4.2. Global existence and a uniform width bound 23
5. Algorithms 26
5.1. Algorithm 1: the log penalty on the normalized measure 27
5.2. Algorithm 2: the fractional-exponent penalty on the sphere 30
6. Numerical examples 31
6.1. Van der Pol Oscillator 31
6.2. Pendulum Swing-Up 36
6.3. Allen–Cahn equation 40

7. Conclusion and outlook 43
References 45

1. Introduction

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$
(P)\qquad
\begin{aligned}
&\inf_{u\in L^2(t_0,T;\mathbb{R}^{m_u})} \mathcal{C}(u;t_0,x) := \int_{t_0}^{T} \big(h(y(t)) + \eta|u(t)|^2\big)\,dt, \quad \eta>0,\\
&\text{s.t.}\\
&\dot{y} = f(y) + g(y)u,\\
&y(t_0) = x,
\end{aligned}
$$

We assume $y(t)\in\mathbb{R}^d$, $u(t)\in\mathbb{R}^{m_u}$, $h\in C^1(\mathbb{R}^d)$, $f\in C^1(\mathbb{R}^d;\mathbb{R}^d)$, and $g\in C^1(\mathbb{R}^d;\mathbb{R}^{d\times m_u})$. We assume a minimizer exists.

One approach computes an open-loop trajectory of the state, costate, and control for a fixed initial condition x. The Pontryagin maximum principle (PMP) yields a coupled forward–backward two-point boundary value problem:

$$
\text{(TPBVP)}\qquad
\begin{cases}
\dfrac{d}{dt}y^*(t) = f(y^*(t)) + g(y^*(t))u^*(t),\\
y^*(t_0) = x,\\
-\dfrac{d}{dt}p^*(t) = \partial_y\Big(f(y^*(t)) + g(y^*(t))u^*(t)\Big)^\top p^*(t) + \partial_y h(y^*(t)),\\
p^*(T) = 0.
\end{cases}
$$

The stationarity condition with respect to the control gives

$$
u^*(t) = -\frac{1}{2\eta}g(y^*(t))^\top p^*(t) \quad \text{for almost every } t\in(t_0,T).
$$

An open-loop control does not adjust in response to state disturbances [1]. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$
V(t,x) := \inf_u \mathcal{C}(u;t,x)
$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation in the viscosity sense:

$$
\text{(HJB)}\qquad
\begin{cases}
-\partial_t V + \dfrac{1}{4\eta}\nabla_x V(t,x)^\top g(x)g(x)^\top \nabla_x V(t,x) - \nabla_x V(t,x)^\top f(x) - h(x) = 0,\\
V(T,x) = 0.
\end{cases}
$$

At points where V is differentiable, the equation holds classically and Hamiltonian minimization gives the feedback $u^*(t,x) = -(2\eta)^{-1}g(x)^\top\nabla_x V(t,x)$. The value function encodes the optimal cost from every state. State feedback adapts the input to the current state.

For nonlinear control problems, one usually seeks an optimal feedback law, which can be constructed from the value function obtained by solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [19], generates the training data with (TPBVP) and approximates the optimal value function from it. The feedback law is then synthesized from the fitted value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with adjoint information:

a state-adjoint-control triple is generated by solving the open-loop problem, and, for an optimal trajectory, the adjoint supplies the value gradient at differentiability points. The value function is then approximated by hyperbolic cross basis functions with an ℓ^1 penalty, and including the adjoint data targets an H^1 approximation. Neural network approximations form another growing line of work [17, 24]. Other prior information, such as semiconcavity of the value function, can also be incorporated. Alternatively, one can use an HJB- or physics-informed loss [9, 18, 23, 25, 16].

For $\boldsymbol{\omega} = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$\mathcal{N}_{\boldsymbol{\omega},c}(x) := \sum_{n=1}^{N} c_n \rho(a_n \cdot x + b_n) \in \mathbb{R}, \tag{1}$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Throughout, representations of finite networks are understood after repeated locations are merged and zero coefficients are discarded; their width N is the number of remaining terms. Its gradient is

$$\nabla_x \mathcal{N}_{\boldsymbol{\omega},c}(x) = \sum_{n=1}^{N} c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

We write $V(x) := V(0, x)$. The training data are computed numerically from (TPBVP). For the trajectory starting from x^m, let v^m denote its accumulated cost and $g^m := p^{*,m}(0)$ its initial costate. For an exact optimal trajectory at a sample where V is differentiable, sensitivity relations give $g^m = \nabla_x V(x^m)$ [8, 6, 7]. At a nondifferentiable point, competing optimal paths may supply different costates; no classical gradient is asserted there. Thus the data consist of triples $\{x^m, v^m, g^m\}_{m=1}^M$. We have the minimization problem

$$\min_{N,\boldsymbol{\omega},c} l^M(\mu_{\boldsymbol{\omega},c}), \qquad l^M(\mu_{\boldsymbol{\omega},c}) := \frac{1}{2M} \sum_{m=1}^{M} \left(|v^m - \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 + |g^m - \nabla \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 \right).$$

Fix a bounded Lipschitz domain $D \subset \mathbb{R}^d$, Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D.

Another focus of this study is the sparsity of the network. The number of neurons N is itself an optimization variable. Although width alone does not generally explain a model's inductive bias [20], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is useful for certain applications, such as knowledge distillation and deconvolution of point-like objects.

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [4, 2]. Pieper and Petrosyan [21] showed that convex penalties cannot effectively eliminate redundant neurons when they cluster in a small neighborhood and introduced a nonconvex penalty for this purpose. However this formulation poses difficulties for gradient fitting since the formulation is restricted to the ReLU setting: the second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate higher-order derivatives. We need to extend to smooth activation functions.

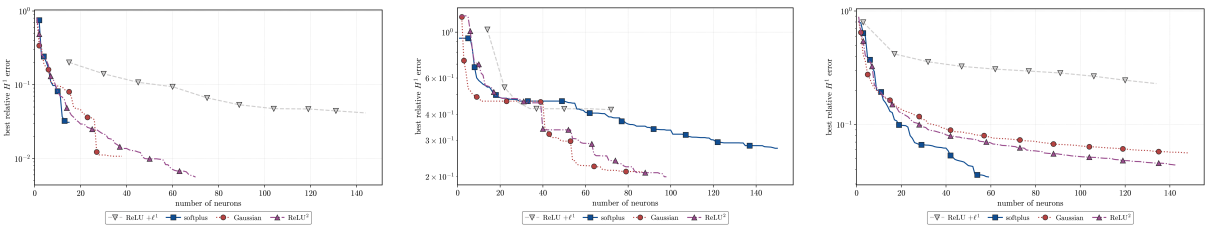
Such an extension requires a treatment of the unbounded space of inner parameters for gradient training. 1. For positively k-homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. However, for $k \geq 2$, the induced nonconvex penalty has infinite right derivative at zero and the usual functional tools to prove the existence and optimality are not applicable. 2. For nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes the atom and cannot be removed by normalization. Without normalization, sequences with $|\omega| \to \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We address those difficulties in each case.

Contributions. The contributions are the following.

(1) *Nonhomogeneous activations on the unbounded parameter domain.* We adopt the objective function with gradient information from [1] and the nonconvex penalty from [21]. Like in [21], we require the concave scalar penalties to have finite positive right slope. However, unlike the original paper which focuses on the ReLU case, we allow the activation functions to be nonhomogeneous, with inner parameters in an unbounded domain. We prove a weak representation of the penalty Φ_ϕ and derive weak-* lower semicontinuity (Theorem 3.9 and Corollary 3.10). In Theorem 3.13 with the normalized atom K_p, we extend the existence result of a global minimizer from the value loss with ReLU activation function to the gradient-augmented loss with any continuous activation $\rho : \mathbb{R} \to \mathbb{R}$ satisfying the stated polynomial growth condition, with inner parameters in $\Omega = \mathbb{R}^{d+1}$ (Theorem 3.16). Under strong concavity of the penalty on every bounded interval, we prove that every local minimizer is finite atomic as in [21]. However, we further extend the result and prove that the number of neurons of optimal network admits an a priori bound (Theorem 3.20, Corollary 3.21). We propose a new version of the PDAP algorithm. For the single-atom PDAP scheme with exact search and monotone correction, we prove an $O(n^{-1/2})$ best-iterate bound on the first-order optimality residual for atom insertion (Theorem 5.1, Corollary 5.2).

(2) *The k-homogeneous formulation.* For positively k-homogeneous activations, we show that quadratic regularization of the inner and outer parameters is equivalent to a penalty proportional to $|c|^{2/(k+1)}$ on the outer coefficients with unit-norm inner parameters (Lemma 4.4), which is extended from the ReLU rescaling equivalence of [21, Appendix F, Proposition 3]. For ReLU^k with $k \geq 2$, the resulting problem is posed over all of $\mathcal{M}(\mathbb{S}^d)$, which has the same infimum as the problem over finite networks (Lemma 4.7). We prove that a global minimizer exists, and every global minimizer is a finite network obeying an explicit a priori width bound (Theorem 4.10). We derive the atomwise optimality conditions and a positive lower bound on the weights of every local minimizer with respect to the total variation norm (Theorem 4.8). Every local minimizer has finite support (Corollary 4.12). An insertion algorithm is proposed based on the optimality condition (Algorithm 2).

Figure 1 shows that the proposed algorithms can achieve lower relative H^1 validation errors with fewer neurons than the traditional ℓ^1-regularized ReLU network on all three benchmarks.



(A) Van der Pol. (B) Pendulum swing-up. (C) Allen–Cahn.

FIGURE 1. Best observed relative H^1 validation error at each support budget for (a) Van der Pol, (b) Pendulum swing-up, and (c) Allen–Cahn. See Section 6 for the experimental settings.

2. PRELIMINARIES

This section reviews the measure-theoretic facts used in the paper. The learning objectives and assumptions are introduced in Section 3.

2.1. **Finite signed Radon measures.** Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω, equipped with the total variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_\Omega f \, d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\operatorname{atom} \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \operatorname{atom} \mu} \mu(\{\omega\})\, \delta_\omega$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}}$ [5]. This gives the decomposition $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$. The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \operatorname{sign}(\mu)\, |\mu|, \qquad \operatorname{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu|\text{-almost everywhere.}$$

2.2. **Measures with a finite moment.** In this subsection let $\Omega = \mathbb{R}^{d_\Omega}$, where $d_\Omega \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$\mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_\Omega w_p \, d|\mu| < \infty \right\}, \qquad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_\Omega w_p \, d|\mu|. \tag{2}$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \qquad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [3, Section 5].

Notation. Throughout the remainder of the paper, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d|_D$ is the population measure, and $L^2(D)$ means $L^2(D, \nu)$. We assume that the population target V belongs to $H^1(D)$. The symbol $M \geq 1$ denotes the number of samples in an empirical loss. The domain of the inner parameters is $\Omega = \mathbb{R}^{d+1}$, and $\rho \in C(\mathbb{R})$ denotes the fixed activation function of (1), with

$$\omega = (a, b), \qquad [K(\omega)](x) := \rho(a \cdot x + b).$$

With this notation, the network in (1) is $\mathcal{N}_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n K(\omega_n)$. The symbol δ_ω denotes the Dirac measure at ω. We reserve p for the moment order.

3. The learning problem

3.1. **The value operator and the gradient-augmented loss.** The finite network $\mathcal{N}_{\boldsymbol{\omega},c}$ corresponds to the finite atomic measure $\mu_{\boldsymbol{\omega},c}$. This motivates the integral formulation in (6), which agrees exactly with the finite sum on atomic measures.

Assumption 3.1 (Regularity and polynomial growth of ρ)**.** The activation satisfies $\rho \in C^1(\mathbb{R})$. There are $C_\rho > 0$, $s_0 \geq 0$, and $s_1 \geq \max\{s_0, 1\}$ such that

(1) $|\rho(z)| \leq C_\rho(1 + |z|)^{s_0}$ for every $z \in \mathbb{R}$;
(2) $|\rho'(z)| \leq C_\rho(1 + |z|)^{s_1 - 1}$ for every $z \in \mathbb{R}$.

Since D is bounded, Assumption 3.1 gives constants $C_D, C_K > 0$ such that

$$\|K(\omega)\|_{L^2(D)} \le C_D(1+|\omega|)^{s_0}. \tag{3}$$

Moreover, $\nabla K(a,b)(x) = a\rho'(a\cdot x+b)$, and hence

$$\|K(\omega)\|_{H^1(D)} \le C_K(1+|\omega|)^{s_1}. \tag{4}$$

If $\omega_n \to \omega$, their preactivations lie in a common compact interval on D. Uniform continuity of ρ and ρ' on that interval, together with $a_n \to a$, therefore gives $K(\omega_n) \to K(\omega)$ in $H^1(D)$. Thus $K : \Omega \to H^1(D)$ is continuous and, in particular, strongly measurable by Pettis.

For $\mu \in \mathcal{M}(\Omega)$ satisfying

$$\int_\Omega \|K(\omega)\|_{H^1(D)}\, d|\mu|(\omega) < \infty, \tag{5}$$

define the $H^1(D)$-valued Bochner integral

$$\mathcal{N}\mu := \int_\Omega K(\omega)\, d\mu(\omega). \tag{6}$$

By (4), condition (5) holds for $\mu \in \mathcal{M}_p(\Omega)$ when $p > s_1$. For $\psi \in C_c^\infty(D)$ and $i \in \{1,\dots,d\}$, the growth bound permits Fubini's theorem and gives

$$\begin{aligned}\int_D (\mathcal{N}\mu)\,\partial_i\psi\,dx &= \int_\Omega \int_D K(\omega)\,\partial_i\psi\,dx\,d\mu(\omega)\\ &= -\int_D \left(\int_\Omega \partial_i K(\omega)\,d\mu(\omega)\right)\psi\,dx.\end{aligned}$$

Consequently,

$$\nabla(\mathcal{N}\mu) = \int_\Omega \nabla K(\omega)\,d\mu(\omega) \quad \text{in } L^2(D;\mathbb{R}^d).$$

For measures satisfying only $\int_\Omega \|K(\omega)\|_{L^2(D)}\, d|\mu|(\omega) < \infty$, the integral in (6) is well defined as an $L^2(D)$-valued Bochner integral. By (3), this includes $\mathcal{M}_p(\Omega)$ when $p > s_0$. In particular, $\mathcal{N}\mu_{\boldsymbol{\omega},c} = \mathcal{N}_{\boldsymbol{\omega},c}$.

Whenever $\mathcal{N}\mu \in H^1(D)$, set $r_\mu := \mathcal{N}\mu - V$. If the $L^2(D)$-valued integral defining $\mathcal{N}\mu$ does not exist, we set $L(\mu) = +\infty$:

$$L(\mu) := \begin{cases} \dfrac{1}{2}\|\mathcal{N}\mu - V\|^2_{L^2(D)} + \dfrac{1}{2}\|\nabla(\mathcal{N}\mu) - \nabla_x V\|^2_{L^2(D)}, & \mathcal{N}\mu \in H^1(D),\\ +\infty, & \text{otherwise.}\end{cases}$$

For sampled data $(v^m, g^m)_{m=1}^M$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$l^M(\mu) := \frac{1}{2M}\sum_{m=1}^M \left(|\mathcal{N}\mu(x^m) - v^m|^2 + |\nabla\mathcal{N}\mu(x^m) - g^m|^2\right). \tag{7}$$

3.2. **The nonconvex functional.** Let $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The finite right derivative

$$L_\phi := \phi'(0+)$$

exists and satisfies $0 < L_\phi < \infty$. We use the convention $\phi'(0) := L_\phi$.

Assumption 3.3. The function ϕ is concave, nondecreasing, and coercive, with $\phi(0) = 0$, and belongs to $C^1((0,\infty))$.

Assumptions 3.2 and 3.3 imply that $\phi'(t) > 0$ for every $t > 0$. Indeed, if $\phi'(t_0) = 0$ at some $t_0 > 0$, concavity and monotonicity would give $\phi'(t) = 0$ for every $t \ge t_0$, making ϕ constant there and contradicting coercivity. Hence ϕ is continuous and strictly increasing from $[0,\infty)$ onto $[0,\infty)$. We denote its inverse by ϕ^{-1}.

Assumption 3.4 (Local curvature)**.** The function ϕ belongs to $C^2((0,\infty))$ and satisfies both of the following conditions:

(1) there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;

(2) for every $A > 0$,

$$\gamma_A := -\sup_{0<t\leq A} \phi''(t) > 0.$$

Remark 3.5. The lower curvature control in Assumption 3.4(1) is local and does not follow from global concavity. For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$. Choose $0 < \delta < 1$, retain this formula on $[0,\delta]$, and use its tangent line at δ for $z > \delta$. The resulting function is nondecreasing, concave, and coercive.

The following result follows from standard concavity estimates; see [21].

Lemma 3.6. *Let $\phi : [0,\infty) \to [0,\infty)$ be concave with $\phi(0) = 0$. Then*

(1) *$\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$.*

If in addition $L_\phi := \phi'(0+)$ is finite, then

(2) *$\phi(z) \leq L_\phi z$ for all $z \geq 0$;*

(3) *$\phi(z)/z \longrightarrow L_\phi$ as $z \downarrow 0$;*

(4) *for every nonnegative sequence $(t_j)_{j\geq 1}$ with $\sum_{j\geq 1} t_j < \infty$,*

$$\phi\left(\sum_{j\geq 1} t_j\right) \leq \sum_{j\geq 1} \phi(t_j).$$

Proof. Concavity and $\phi(0) = 0$ make $z \mapsto \phi(z)/z$ nonincreasing on $(0,\infty)$: for $0 < z_1 \leq z_2$, writing $z_1 = \frac{z_1}{z_2}z_2 + \left(1 - \frac{z_1}{z_2}\right)\cdot 0$ and using concavity together with $\phi(0) = 0$ gives $\phi(z_1) \geq \frac{z_1}{z_2}\phi(z_2)$.

(1) Suppose without loss of generality that $0 < z_1 \leq z_2$. Then

$$\begin{aligned}
\phi(z_1 + z_2) &\leq \frac{z_1 + z_2}{z_2}\phi(z_2) && \text{because } \phi(z)/z \text{ is nonincreasing} \\
&= \phi(z_2) + z_1\frac{\phi(z_2)}{z_2} \\
&\leq \phi(z_2) + \phi(z_1) && \text{by the same monotonicity argument.}
\end{aligned}$$

The cases with $z_1 = 0$ or $z_2 = 0$ follow from $\phi(0) = 0$.

(2) and (3) By the same monotonicity argument and $\phi(0) = 0$,

$$\frac{\phi(z)}{z} \leq \lim_{s\downarrow 0}\frac{\phi(s)}{s} = \lim_{s\downarrow 0}\frac{\phi(s) - \phi(0)}{s} = L_\phi \qquad (z > 0),$$

and the displayed limit is precisely (3).

(4) By (2) the function ϕ is continuous at 0, and a finite concave function is continuous on $(0,\infty)$. Iterating (1) over partial sums gives

$$\phi\Big(\sum_{j\leq N} t_j\Big) \leq \sum_{j\leq N}\phi(t_j) \leq \sum_{j\geq 1}\phi(t_j),$$

the last step because the terms are nonnegative. Letting $N \to \infty$ and using continuity of ϕ at $\sum_{j\geq 1} t_j$ proves the claim. □

Assumption 3.3 implies the standing hypotheses of Lemma 3.6, and Assumption 3.2 supplies the finite right slope needed for parts (2)–(4). Neither coercivity nor differentiability on $(0,\infty)$ is used, so the lemma also applies under the weaker hypotheses of Theorem 3.9.

Proposition 3.7 (Second-order bounds)**.** *Suppose that $\phi(0) = 0$ and Assumption 3.2 holds.*

(1) *If the derivative bound in Assumption 3.4(1) holds, then $\phi(z) \geq L_\phi z - \frac{\gamma_1}{2}z^2$ for $0 \leq z \leq \hat{z}_1$;*

(2) *if* $\phi \in C^2((0,\infty))$ *and Assumption 3.4(2) holds, then, for every* $A > 0$, $\phi(z) \le L_\phi z - \frac{\gamma_A}{2} z^2$ *for* $0 \le z \le A$.

Proof. For $\varepsilon > 0$, apply the fundamental theorem of calculus on $[\varepsilon, z]$ and then let $\varepsilon \downarrow 0$. The definition of the finite right derivative and $\phi(0) = 0$ give

$$\phi(z) - L_\phi z = \int_0^z [\phi'(s) - L_\phi]\, ds.$$

The first conclusion follows from Assumption 3.4(1); see also [21].

For the second, Assumption 3.4(2) gives $\phi''(t) \le -\gamma_A$ on $(0, A]$. Integrating this bound with respect to t from ε to s gives $\phi'(s) - \phi'(\varepsilon) \le -\gamma_A(s - \varepsilon)$. Letting $\varepsilon \downarrow 0$ and using $\phi'(\varepsilon) \to L_\phi$, we obtain $\phi'(s) - L_\phi \le -\gamma_A s$ for $0 < s \le A$. Integrating this inequality with respect to s from 0 to z proves the claim. □

For an open set $U \subset \Omega$, let $C_c(U)$ denote the continuous functions with compact support in U. Define

$$\Phi_\phi(\mu) := L_\phi \|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \operatorname{atom} \mu} \phi\big(|\mu(\{\omega\})|\big), \qquad \mu \in \mathcal{M}(\Omega). \tag{8}$$

We give a self-contained representation and proof of weak-* lower semicontinuity for this specific nonconvex functional under weaker assumptions. Another proof for the general setting can be found in Bouchitté–Buttazzo [5, Theorem 3.3 and Remark 3.5].

Lemma 3.8. *Let* $U \subset \Omega = \mathbb{R}^{d+1}$ *be open, let* $\mu \in \mathcal{M}(U)$, *and let* $\delta > 0$. *Suppose that*

$$|\mu(\{\omega\})| \le \frac{\delta}{4} \qquad \text{for every } \omega \in U.$$

Then, for every $\varepsilon > 0$, *there are* $m \ge 1$ *and functions* $h_1, \dots, h_m \in C_c(U)$ *with pairwise disjoint supports and* $|h_i| \le 1$ *such that*

$$\left| \int_U h_i \, d\mu \right| \le \delta \quad (i = 1, \dots, m), \qquad \sum_{i=1}^m \left| \int_U h_i \, d\mu \right| \ge \|\mu\|_{\mathcal{M}(U)} - \varepsilon.$$

Proof. If $\varepsilon \ge \|\mu\|_{\mathcal{M}(U)}$, take $m = 1$ and $h_1 = 0$. Henceforth assume $0 < \varepsilon < \|\mu\|_{\mathcal{M}(U)}$. Let $U = P \cup N$ be a Hahn decomposition of μ. By inner regularity, there are disjoint compact sets $K_+ \subset P$ and $K_- \subset N$ such that

$$|\mu|\big(U \setminus (K_+ \cup K_-)\big) < \frac{\varepsilon}{4}. \tag{9}$$

We now construct a finite partition into sets of $|\mu|$-measure less than $\delta/2$, including both the atomic and nonatomic parts. For every $\omega \in K_+ \cup K_-$, continuity from above gives

$$|\mu|\big(B_r(\omega)\big) \longrightarrow |\mu|(\{\omega\}) \le \frac{\delta}{4} \qquad (r \downarrow 0).$$

Choose $r_\omega > 0$ such that $|\mu|(B_{r_\omega}(\omega)) < \delta/2$. Compactness gives a finite subcover. Let $B_1, \dots, B_N$ be these balls. For each $\sigma \in \{+, -\}$, define inductively

$$E_i^\sigma := (K_\sigma \cap B_i) \setminus \bigcup_{j<i} (K_\sigma \cap B_j), \qquad i = 1, \dots, N,$$

where the union is empty for $i = 1$. After discarding empty sets, the E_i^σ form a pairwise disjoint Borel partition of K_σ, with $E_i^\sigma \subset B_i$. A finite Borel partition $E_1, \dots, E_{m_0}$ for some $m_0 \ge 1$ of $K_+ \cup K_-$ is constructed such that each E_i lies in either P or N and $|\mu|(E_i) < \delta/2$.

By inner regularity, choose compact sets $C_i \subset E_i$ whose total omitted mass is less than $\varepsilon/4$, and discard the empty sets. At least one set remains because $\varepsilon < \|\mu\|_{\mathcal{M}(U)}$ and (9). Relabel the

remaining sets $C_1, \dots, C_m$. They are pairwise disjoint and satisfy

$$\sum_{i=1}^{m} |\mu|(C_i) \geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4}, \qquad |\mu|(C_i) \leq \frac{\delta}{2}. \tag{10}$$

Each C_i lies entirely in either P or N.

Because the compact sets C_i are finitely many, pairwise disjoint, and contained in the open set U, they have open neighborhoods with pairwise disjoint compact closures contained in U. Outer regularity lets us shrink these neighborhoods, while keeping each C_i inside, to obtain open sets U_i satisfying

$$\sum_{i=1}^{m} |\mu|(U_i \setminus C_i) < \min\left\{\frac{\varepsilon}{4}, \frac{\delta}{2}\right\}. \tag{11}$$

For each i, the continuous function

$$\chi_i(\omega) := \frac{\operatorname{dist}(\omega, U_i^c)}{\operatorname{dist}(\omega, C_i) + \operatorname{dist}(\omega, U_i^c)}$$

equals one on C_i, vanishes outside U_i, and has compact support in U. Set $h_i = \chi_i$ when $C_i \subset P$ and $h_i = -\chi_i$ when $C_i \subset N$. Their supports are pairwise disjoint and $|h_i| \leq 1$. Moreover,

$$\left|\int_U h_i \, d\mu\right| \leq |\mu|(U_i) \leq |\mu|(C_i) + |\mu|(U_i \setminus C_i) \leq \delta, \qquad \text{by (10)–(11)},$$

and

$$\begin{aligned}
\sum_{i=1}^{m} \left|\int_U h_i \, d\mu\right| &\geq \sum_{i=1}^{m} \big(|\mu|(C_i) - |\mu|(U_i \setminus C_i)\big) \\
&\geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (10)–(11)}, \\
&\geq \|\mu\|_{\mathcal{M}(U)} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (9)}, \\
&= \|\mu\|_{\mathcal{M}(U)} - \frac{3\varepsilon}{4}.
\end{aligned}$$

This is stronger than the required lower bound. □

Theorem 3.9 (Representation formula for the nonconvex functional)**.** *Suppose that $\phi : [0,\infty) \to [0,\infty)$ is nondecreasing and concave, $\phi(0) = 0$, and*

$$L_\phi := \phi'(0+) \in (0, \infty).$$

Then, for every $\mu \in \mathcal{M}(\Omega)$,

$$\Phi_\phi(\mu) = \sup\left\{\sum_{i=1}^{m} \phi\left(\left|\int_\Omega f_i \, d\mu\right|\right) : \begin{array}{l} m \geq 1, \quad f_i \in C_c(\Omega), \quad |f_i| \leq 1, \\ \operatorname{supp}(f_i) \text{ are pairwise disjoint} \end{array}\right\}. \tag{12}$$

Proof. We prove the representation by first bounding every admissible finite sum from above and then constructing sums that approach $\Phi_\phi(\mu)$ from below. The hypotheses are exactly the standing hypotheses of Lemma 3.6 together with $L_\phi \in (0,\infty)$, so parts (1)–(3) of that lemma give

$$\phi(r+s) \leq \phi(r) + \phi(s), \qquad \phi(r) \leq L_\phi r, \qquad \frac{\phi(r)}{r} \longrightarrow L_\phi \quad (r \downarrow 0), \tag{13}$$

and part (4) gives the countable form $\phi\big(\sum_{j\geq 1} t_j\big) \leq \sum_{j\geq 1} \phi(t_j)$.

Write the atomic part of μ as

$$\mu_{\text{atom}} = \sum_{j\geq 1} c_j \delta_{\omega_j},$$

where the ω_j are distinct and $c_j \neq 0$. For an admissible family $(f_i)_{i=1}^m$, set

$$A_i := \sum_{j\geq 1} |c_j|\,|f_i(\omega_j)|, \qquad B_i := \int_\Omega |f_i|\,d|\mu_{\mathrm{cont}}|.$$

$$\begin{aligned}
\phi\left(\left|\int_\Omega f_i\,d\mu\right|\right) &\leq \phi(A_i + B_i) && \text{by the triangle inequality}\\
&\leq \sum_{\omega_j\in\mathrm{supp}(f_i)} \phi(|c_j|) + L_\phi B_i && \text{by Lemma 3.6(2), (4) and } |f_i|\leq 1.
\end{aligned}$$

The disjoint supports count each atom at most once and satisfy $\sum_i |f_i| \leq 1$. Therefore

$$\begin{aligned}
\sum_{i=1}^m \phi\left(\left|\int_\Omega f_i\,d\mu\right|\right) &\leq \sum_{j\geq 1}\phi(|c_j|) + L_\phi \int_\Omega \sum_{i=1}^m |f_i|\,d|\mu_{\mathrm{cont}}|\\
&\leq \sum_{j\geq 1}\phi(|c_j|) + L_\phi \|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} = \Phi_\phi(\mu).
\end{aligned}$$

This proves the upper bound in (12).

For the reverse inequality, the claim is immediate when $\mu = 0$. Otherwise, fix $\varepsilon > 0$. Choose $\theta \in (0, L_\phi)$ such that $\theta\|\mu\|_{\mathcal{M}(\Omega)} < \varepsilon/4$. By the last limit in (13), choose $\delta > 0$ such that

$$\phi(r) \geq (L_\phi - \theta)r \qquad (0 \leq r \leq \delta). \tag{14}$$

Let

$$F := \{j \geq 1 : |c_j| > \delta/8\}, \qquad \mu_{\mathrm{rem}} := \mu - \sum_{j\in F} c_j \delta_{\omega_j}.$$

The set F is finite and every point mass of μ_{rem} has magnitude at most $\delta/8$.

Because F is finite, its points have pairwise disjoint relatively compact neighborhoods. Moreover, $\mu_{\mathrm{rem}}(\{\omega_j\}) = 0$ for $j \in F$, so continuity of the measure from above allows these neighborhoods to be chosen with arbitrarily small total $|\mu_{\mathrm{rem}}|$-mass. Using the construction in Lemma 3.8, choose $g_j \in C_c(\Omega)$, $j \in F$, with pairwise disjoint supports, $|g_j| \leq 1$, and

$$g_j(\omega_j) = \mathrm{sign}(c_j), \qquad \sum_{j\in F} |\mu_{\mathrm{rem}}|(\mathrm{supp}(g_j)) < \tau,$$

where $\tau > 0$ can be chosen arbitrarily small. For every $j \in F$,

$$\left|\left|\int_\Omega g_j\,d\mu\right| - |c_j|\right| \leq |\mu_{\mathrm{rem}}|(\mathrm{supp}(g_j)).$$

By continuity of ϕ and finiteness of F, first choose τ sufficiently small and then choose the supports so that

$$\sum_{j\in F}\phi\left(\left|\int_\Omega g_j\,d\mu\right|\right) \geq \sum_{j\in F}\phi(|c_j|) - \frac{\varepsilon}{4}, \qquad 2L_\phi\tau < \frac{\varepsilon}{4}. \tag{15}$$

Set

$$U := \Omega \setminus \bigcup_{j\in F} \mathrm{supp}(g_j) \qquad \text{and} \qquad \lambda := \mu_{\mathrm{rem}}|_U.$$

Lemma 3.8 applied on the open set U gives $h_1, \ldots, h_m \in C_c(U)$ whose zero extensions belong to $C_c(\Omega)$, have pairwise disjoint supports, and satisfy

$$\left|\int_U h_i\,d\lambda\right| \leq \delta, \qquad \sum_{i=1}^m \left|\int_U h_i\,d\lambda\right| \geq \|\lambda\|_{\mathcal{M}(U)} - \tau \geq \|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau.$$

Together with the g_j, the zero extensions form an admissible family in (12). Since every h_i is supported in U and $\mu|_U = \lambda$, its integrals against μ and λ agree. Therefore (14) and (15) yield

$$\begin{aligned}
&\sum_{j\in F} \phi\left(\left|\int_\Omega g_j \, d\mu\right|\right) + \sum_{i=1}^{m} \phi\left(\left|\int_\Omega h_i \, d\mu\right|\right) \\
&\quad \geq \sum_{j\in F} \phi(|c_j|) - \frac{\varepsilon}{4} + (L_\phi - \theta)\big(\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau\big) \\
&\quad \geq \Phi_\phi(\mu) - \frac{\varepsilon}{4} - \theta\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2L_\phi\tau \qquad \text{by } \Phi_\phi(\mu_{\mathrm{rem}}) \leq L_\phi\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} \\
&\quad > \Phi_\phi(\mu) - \varepsilon.
\end{aligned}$$

Taking the supremum and then letting $\varepsilon \downarrow 0$ proves the reverse inequality. □

Corollary 3.10. *Under the assumptions of Theorem 3.9, the functional Φ_ϕ is weak-* lower semicontinuous on $\mathcal{M}(\Omega) = C_0(\Omega)^*$. If ϕ is also coercive, then every sublevel of Φ_ϕ is bounded in total variation.*

Proof. For each fixed admissible family in (12), weak-* convergence gives convergence of all its integrals, and continuity of ϕ gives convergence of the corresponding finite sum. The representation is a supremum of weak-* continuous functions and is therefore weak-* lower semicontinuous.

For the sublevel bound, apply repeated subadditivity to the first N atoms and let $N \to \infty$. The atom magnitudes are summable, so continuity of ϕ gives

$$\phi\big(\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}\big) \leq \sum_{\omega\in\operatorname{atom}\mu} \phi\big(|\mu(\{\omega\})|\big) \leq \Phi_\phi(\mu), \qquad L_\phi\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} \leq \Phi_\phi(\mu).$$

If $\Phi_\phi(\mu) \leq C$, coercivity bounds $\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}$ in terms of C, while $\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} \leq C/L_\phi$. The atomic part is supported on the countable atom set, while the continuous part assigns that set zero variation. The two parts are therefore mutually singular, so their total variation norms add to $\|\mu\|_{\mathcal{M}(\Omega)}$. Hence the sublevel is bounded in total variation. □

3.3. **The normalized problem.** For $\alpha > 0$, define

$$J_0(\mu) := L(\mu) + \alpha\Phi_\phi(\mu)$$

for the unnormalized problem on the parameter domain.

Escape on the unbounded parameter domain. For positively homogeneous activation functions the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The regularizer Φ_ϕ depends on the sizes of the outer coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \to \infty$, the fidelity need not be lower semicontinuous.

Example 3.11 (Failure of weak-* lower semicontinuity for tanh). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a,b_n)}$ with $b_n \to \infty$. Then $\mu_n \overset{*}{\rightharpoonup} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \to 1$$

strongly in $H^1(D)$. Hence $\liminf_{n\to\infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity can retain a nonzero value contribution when $K(\omega)$ does not tend to zero as $|\omega| \to \infty$.

Note that $\Phi_\phi(\mu_n) = \phi(1)$ is constant along this sequence, so the unnormalized coefficient penalty does not prevent this escape. The defect is that the unnormalized kernel does not vanish at parameter infinity. The normalized atom introduced in Theorem 3.13 does vanish under the corresponding polynomial growth gap.

Example 3.12 (Nonattainment for softplus). For the activation function $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \to \infty$. The measures $\mu_n = \rho(b_n)^{-1}\delta_{(0,b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha\,\phi(1/\rho(b_n)) \to 0$, so $\inf J_0 = 0$ is not attained. Indeed, if $J_0(\mu) = 0$, then $\Phi_\phi(\mu) = 0$. Strict positivity of ϕ on $(0,\infty)$ and $L_\phi > 0$ force both the atomic and continuous parts of μ to vanish. Thus $\mu = 0$, but $L(0) = \frac{1}{2}\|1\|^2_{H^1(D)} > 0$, a contradiction.

We need a penalty that is aware of the location of the atom to prevent the escape. The space of measures with finite p-th moment is the right space to work with. For $p > 0$, let w_p and $\mathcal{M}_p(\Omega)$ be as in (2). For $\mu \in \mathcal{M}_p(\Omega)$, we define the normalized measure and normalized atom by

$$d\mu_p := w_p\, d\mu, \qquad K_p(\omega) := \frac{K(\omega)}{w_p(\omega)}.$$

Theorem 3.13. *Multiplication by w_p is a linear isometric bijection*

$$\mathcal{M}_p(\Omega) \longrightarrow \mathcal{M}(\Omega), \qquad \mu \longmapsto \mu_p,$$

with inverse $d\mu = w_p^{-1} d\mu_p$, and

$$\|\mu\|_{\mathcal{M}_p(\Omega)} = \|\mu_p\|_{\mathcal{M}(\Omega)}. \tag{16}$$

Moreover, μ_n converges weak- to μ in $C_{w_p^{-1}}(\Omega)^*$ if and only if $(\mu_n)_p \overset{*}{\rightharpoonup} \mu_p$ in $C_0(\Omega)^*$ and*

$$(\mu_p)_{\mathrm{atom}} = w_p \mu_{\mathrm{atom}}, \qquad (\mu_p)_{\mathrm{cont}} = w_p \mu_{\mathrm{cont}}. \tag{17}$$

If part (1) of Assumption 3.1 holds and $p > s_0$, then $K_p \in C_0(\Omega; L^2(D))$ and

$$\mathcal{N}\mu = \int_\Omega K\, d\mu = \int_\Omega K_p\, d\mu_p. \tag{18}$$

Finally,

$$\Phi_\phi(\mu_p) = L_\phi \int_\Omega w_p\, d|\mu_{\mathrm{cont}}| + \sum_{\omega \in \mathrm{atom}\,\mu} \phi\big(w_p(\omega)|\mu(\{\omega\})|\big). \tag{19}$$

Proof. The isometric identification with $\mathcal{M}(\Omega)$, including its inverse, the norm identity, and the weak-* equivalence, is the specialization of [3, Section 5.1] to the weight w_p; in particular, this gives (16). Since multiplication by the strictly positive function w_p preserves null sets, the atom set, the polar sign, and nonatomicity, (17) follows immediately.

It remains to treat the network map. The continuity of K_p follows from that of K and w_p. For $r = |\omega| \geq 1$, the growth estimate (3) gives

$$\begin{aligned}\|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} \\ &\leq 2^{s_0} C_D r^{s_0 - p} \longrightarrow 0, \qquad r \to \infty,\end{aligned}$$

because $p > s_0$. Thus $K_p \in C_0(\Omega; L^2(D))$. Using $d\mu_p = w_p\, d\mu$ and $K_p = K/w_p$ in the Bochner integral gives (18). Finally, (17) and $|w_p \mu_{\mathrm{cont}}| = w_p |\mu_{\mathrm{cont}}|$ give (19) directly from (8). □

For $\alpha > 0$, define the adopted objective

$$J(\mu) := L(\mu) + \alpha \Phi_\phi(\mu_p), \qquad \mu \in \mathcal{M}_p(\Omega).$$

For a finite atomic measure $\mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n \delta_{\omega_n}$, this becomes

$$J(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big).$$

Definition 3.14 (Local minimizer)**.** A measure $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if $J(\bar{\mu}) < \infty$ and there is $\varepsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \qquad \text{for all} \qquad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} < \varepsilon.$$

By Theorem 3.13, this is exactly local minimality with respect to the total variation norm in the normalized coordinate μ_p.

3.4. **Existence of a global minimizer.** We apply the direct method in the normalized coordinate. Weak-* compactness controls μ_p, Corollary 3.10 supplies the penalty lower semicontinuity and sublevel bound, and Lemma 3.15 gives the strong convergence needed for the value part of the fidelity.

Lemma 3.15. *Assume (3), let $p > s_0$, and suppose that*

$$(\mu_n)_p \overset{*}{\rightharpoonup} \mu_p \quad \text{in } \mathcal{M}(\Omega).$$

Then $\mathcal{N}\mu_n \to \mathcal{N}\mu$ strongly in $L^2(D)$.

Proof. We use a compactly supported cutoff to pass to the weak-* limit locally and then control the tails uniformly. By Theorem 3.13,

$$\mathcal{N}\mu_n = \int_\Omega K_p \, d(\mu_n)_p, \qquad \mathcal{N}\mu = \int_\Omega K_p \, d\mu_p.$$

Set

$$C := \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)} < \infty,$$

where finiteness follows from weak-* convergence and the uniform boundedness principle. Weak-* lower semicontinuity of total variation gives $\|\mu_p\|_{\mathcal{M}(\Omega)} \le C$.

For $R \ge 1$, choose $\chi_R \in C_c(\Omega)$ with $0 \le \chi_R \le 1$, $\chi_R = 1$ on B_R, and $\operatorname{supp}\chi_R \subset B_{R+1}$. For fixed R and $x \in D$, the function

$$\omega \longmapsto \chi_R(\omega)[K_p(\omega)](x)$$

belongs to $C_c(\Omega)$. Hence weak-* convergence gives

$$\int_\Omega \chi_R(\omega)[K_p(\omega)](x) \, d(\mu_n)_p(\omega) \longrightarrow \int_\Omega \chi_R(\omega)[K_p(\omega)](x) \, d\mu_p(\omega)$$

for every $x \in D$.

Since D is bounded and $\operatorname{supp}\chi_R$ is compact,

$$M_R := \sup_{\omega \in \operatorname{supp}\chi_R,\ x \in D} |[K_p(\omega)](x)| < \infty.$$

The truncated integrals converge for every $x \in D$ and are bounded in absolute value by CM_R. Since D is bounded, the constant function CM_R belongs to $L^2(D)$. Dominated convergence then yields

$$\int_\Omega \chi_R K_p \, d(\mu_n)_p \longrightarrow \int_\Omega \chi_R K_p \, d\mu_p \qquad \text{strongly in } L^2(D).$$

It remains to remove the cutoff. If $r := |\omega| > R \ge 1$, then

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\le C_D \frac{(1+r)^{s_0}}{1+r^p} && \text{by (3)} \\ &\le 2^{s_0} C_D r^{s_0-p} \le 2^{s_0} C_D R^{s_0-p}, && \text{since } p > s_0 \text{ and } r > R. \end{aligned}$$

Consequently,

$$\left\| \int_\Omega (1-\chi_R) K_p \, d(\mu_n)_p \right\|_{L^2(D)} \le 2^{s_0} C_D C R^{s_0-p},$$

$$\left\| \int_\Omega (1-\chi_R) K_p \, d\mu_p \right\|_{L^2(D)} \le 2^{s_0} C_D C R^{s_0-p}.$$

For fixed R, let $n \to \infty$ in the truncated integrals. Then let $R \to \infty$; the two tail bounds vanish because $s_0 - p < 0$. This proves the claim. □

Theorem 3.16 (Existence on the unbounded parameter domain)**.** *Let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let $\rho \in C(\mathbb{R})$ satisfy Assumption 3.1(1). If $p > s_0$ and $\alpha > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. Since $V \in H^1(D)$,

$$J(0) = \frac{1}{2}\|V\|^2_{H^1(D)} =: C < \infty.$$

Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \le C + 1$. The fidelity and penalty are nonnegative, so

$$\Phi_\phi((\mu_n)_p) \le \frac{C+1}{\alpha}.$$

The bound on sublevels in the total variation norm given by Corollary 3.10 shows that $((\mu_n)_p)$ is bounded in $\mathcal{M}(\Omega)$. Weak-* compactness and separability of $C_0(\Omega)$ give, after passing to a subsequence,

$$(\mu_n)_p \overset{*}{\rightharpoonup} \bar{\mu}_p.$$

Theorem 3.13 defines a unique $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with normalized coordinate $\bar{\mu}_p$. Lemma 3.15 gives

$$\mathcal{N}\mu_n \longrightarrow \mathcal{N}\bar{\mu} \qquad \text{strongly in } L^2(D). \tag{20}$$

The fidelity bound gives

$$\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \le \|\nabla_x V\|_{L^2(D)} + \sqrt{2(C+1)}.$$

After another subsequence, we have $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$ for some g. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_{n\to\infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = -\lim_{n\to\infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = -\int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi \, dx,$$

the last step by (20). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The preceding argument shows that the gradients converge weakly in $L^2(D)^d$ to $\nabla(\mathcal{N}\bar{\mu})$. By weak lower semicontinuity of the squared L^2 norm,

$$\frac{1}{2}\|\nabla(\mathcal{N}\bar{\mu}) - \nabla_x V\|^2_{L^2(D)} \le \liminf_{n\to\infty} \frac{1}{2}\|\nabla(\mathcal{N}\mu_n) - \nabla_x V\|^2_{L^2(D)}.$$

Thus the gradient part of the fidelity is lower semicontinuous along the chosen weak-* convergent subsequence of normalized measures. Together with the strong L^2 convergence of the values in (20), this yields

$$L(\bar{\mu}) \le \liminf_{n\to\infty} L(\mu_n).$$

Corollary 3.10 gives

$$\Phi_\phi(\bar{\mu}_p) \le \liminf_{n\to\infty} \Phi_\phi((\mu_n)_p).$$

Combining the two inequalities proves $J(\bar{\mu}) \le \liminf_n J(\mu_n) = \inf J$. □

3.5. **Optimality conditions and finite support.** Let $\bar{\mu}$ be a local minimizer. Under Assumption 3.1 and $p > s_1$, the Gâteaux derivative at $\mu \in \mathcal{M}_p(\Omega)$ in any direction $\mu' \in \mathcal{M}_p(\Omega)$ is

$$DL(\mu)[\mu'] := \lim_{t\to 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_\mu, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

The corresponding derivative profile is

$$P_\mu(\omega) := DL(\mu)[\delta_\omega] = \langle r_\mu, K(\omega) \rangle_{H^1(D)}.$$

Define its normalized form and the uniform normalized bound by

$$P_{p,\mu}(\omega) := \frac{P_\mu(\omega)}{w_p(\omega)} = \langle r_\mu, K_p(\omega) \rangle_{H^1(D)}, \qquad B_p := \sup_{\omega\in\Omega} \|K_p(\omega)\|_{H^1(D)}.$$

At the local minimizer write $\bar{P} := P_{\bar{\mu}}$ and $\bar{P}_p := P_{p,\bar{\mu}}$. Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ,

$$\bar{P}(a,b) = \int_D r_{\bar{\mu}}(x)\rho(a\cdot x+b)\,dx + \int_D \nabla r_{\bar{\mu}}(x)\cdot a\rho'(a\cdot x+b)\,dx.$$

The second term carries the factor a, so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded. By Cauchy–Schwarz, (4) gives

$$|\bar{P}(\omega)| \le \|r_{\bar{\mu}}\|_{H^1(D)}\|K(\omega)\|_{H^1(D)} \le C_P(1+|\omega|)^{s_1}, \qquad C_P := C_K\|r_{\bar{\mu}}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω. If $p > s_1$, then

$$K_p \in C_0(\Omega; H^1(D)), \qquad \bar{P}_p \in C_0(\Omega), \qquad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar{\mu})[\mu'] = \int_\Omega \bar{P}\,d\mu' = \int_\Omega \bar{P}_p\,d\mu'_p.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

The following theorem adapts the optimality conditions of Pieper and Petrosyan [21] to the normalized measure and the gradient-augmented loss.

Theorem 3.17 (The optimality condition)**.** *Let $\phi : [0,\infty) \to [0,\infty)$ be concave and nondecreasing, with $\phi(0) = 0$, $\phi \in C^1((0,\infty))$, and $0 < L_\phi := \phi'(0+) < \infty$. Let $\bar{\mu} \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, and suppose that Assumption 3.1 holds and $p > s_1$. Then*

$$|\bar{P}_p(\omega)| \le \alpha L_\phi \qquad \text{for every } \omega \in \Omega, \tag{21}$$

$$\bar{P}_p(\omega) = -\alpha\phi'\big(w_p(\omega)|\bar{\mu}(\{\omega\})|\big)\operatorname{sign}\big(\bar{\mu}(\{\omega\})\big) \qquad \text{for } \omega \in \operatorname{atom}\bar{\mu}. \tag{22}$$

Moreover,

$$\bar{P}_p = -\alpha L_\phi \operatorname{sign}(\bar{\mu}_{\mathrm{cont}}) \qquad |(\bar{\mu}_p)_{\mathrm{cont}}|\text{-almost everywhere.} \tag{23}$$

Proof. We use Theorem 3.13 to work with $\bar{\mu}_p$ in the total variation norm. The normalized fidelity derivative in the direction $\xi \in \mathcal{M}(\Omega)$ is

$$\left\langle r_{\bar{\mu}}, \int_\Omega K_p\,d\xi \right\rangle_{H^1(D)} = \int_\Omega \bar{P}_p\,d\xi.$$

Fix $\omega \in \operatorname{atom}\bar{\mu}$ and put $z := \bar{\mu}_p(\{\omega\}) = w_p(\omega)\bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1,1\}$ and sufficiently small $t > 0$, the sign of $z + ts$ equals the sign of z. Since $\bar{\mu}$ is a local minimizer,

$$\begin{aligned} 0 &\le J\left(\bar{\mu} + \frac{ts}{w_p(\omega)}\delta_\omega\right) - J(\bar{\mu}) \\ &= ts\bar{P}_p(\omega) + \frac{t^2}{2}\|K_p(\omega)\|^2_{H^1(D)} + \alpha\big[\phi(|z+ts|) - \phi(|z|)\big]. \end{aligned}$$

Dividing by t and letting $t \downarrow 0$ gives

$$0 \le s\left[\bar{P}_p(\omega) + \alpha\phi'(|z|)\operatorname{sign}(z)\right].$$

Using both choices of s makes the bracket vanish, and hence

$$\bar{P}_p(\omega) = -\alpha\phi'(|z|)\operatorname{sign}(z).$$

Because $w_p > 0$, $\operatorname{sign}(z) = \operatorname{sign}(\bar{\mu}(\{\omega\}))$, so this is (22). Concavity and monotonicity give $0 \le \phi'(|z|) \le L_\phi$, and hence $|\bar{P}_p(\omega)| \le \alpha L_\phi$ at every atom.

If $\omega \notin \operatorname{atom}\bar{\mu}$, insertion of $ts\delta_\omega$ into the normalized measure changes the penalty by $\phi(t)$. Thus, for each $s \in \{-1,1\}$ and sufficiently small $t > 0$,

$$0 \le J\left(\bar{\mu} + \frac{ts}{w_p(\omega)}\delta_\omega\right) - J(\bar{\mu})$$

$$= ts\bar{P}_p(\omega) + \frac{t^2}{2}\|K_p(\omega)\|^2_{H^1(D)} + \alpha\phi(t).$$

Dividing by t and letting $t \downarrow 0$ yields

$$0 \le s\bar{P}_p(\omega) + \alpha L_\phi.$$

The two choices of s prove (21) away from the atoms, and the preceding atom estimate completes its proof.

It remains to identify the continuous part. Set $\xi := (\bar{\mu}_p)_{\mathrm{cont}}$. Since $K_p \in C_0(\Omega; H^1(D))$, the directions $\pm\xi$ are admissible. For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu}_p + ts\xi$ multiplies the continuous part by $1 + ts > 0$. Its one-sided directional derivative therefore gives

$$0 \le s\left[\int_\Omega \bar{P}_p \, d\xi + \alpha L_\phi \|\xi\|_{\mathcal{M}(\Omega)}\right].$$

Using both signs shows that the bracket vanishes. On the other hand, (21) makes

$$\bar{P}_p \operatorname{sign}(\xi) + \alpha L_\phi \ge 0$$

$|\xi|$-almost everywhere. Its integral is zero, so the integrand vanishes $|\xi|$-almost everywhere. The sign preservation in Theorem 3.13 gives (23). □

Remark 3.18. Under Assumption 3.1 and $p > s_1$, let $\mu \in \mathcal{M}_p(\Omega)$. Since $P_{p,\mu} \in C_0(\Omega)$, all points satisfying $|P_{p,\mu}(\omega)| > \alpha L_\phi$ lie in the compact set $\{\omega \in \Omega : |P_{p,\mu}(\omega)| \ge \alpha L_\phi\}$, even when P_μ does not decay at parameter infinity. Lemma 3.19 uses a lower superlevel to localize the entire support.

Lemma 3.19. *Let ϕ satisfy Assumptions 3.2 and 3.3, let $p > s_1$, and suppose that Assumption 3.1 holds. Suppose that $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J such that $0 < J(\bar{\mu}) < \infty$. Define*

$$T(\bar{\mu}) := \phi^{-1}\left(\frac{J(\bar{\mu})}{\alpha}\right)$$

and

$$\mathcal{K}(\bar{\mu}) := \left\{\omega \in \Omega : |\bar{P}_p(\omega)| \ge \alpha\phi'\big(T(\bar{\mu})\big)\right\}.$$

Then $\mathcal{K}(\bar{\mu})$ is compact and

$$\operatorname{supp}\bar{\mu}_{\mathrm{atom}} \cup \operatorname{supp}\bar{\mu}_{\mathrm{cont}} \subset \mathcal{K}(\bar{\mu}). \tag{24}$$

Moreover, every atom $\omega \in \operatorname{atom}\bar{\mu}$ satisfies

$$w_p(\omega)|\bar{\mu}(\{\omega\})| \le T(\bar{\mu}).$$

Proof. We use Theorem 3.17 and the fact that $\bar{P}_p$ vanishes at infinity to prove the localization. Concavity and monotonicity make ϕ' positive and nonincreasing, as observed after Assumption 3.3. Since $\bar{P}_p \in C_0(\Omega)$, its closed superlevel set at the strictly positive level $\alpha\phi'(T(\bar{\mu}))$ is compact. This proves the first assertion.

Let $\omega \in \operatorname{atom}\bar{\mu}$ and put

$$t_\omega := |\bar{\mu}_p(\{\omega\})| = w_p(\omega)|\bar{\mu}(\{\omega\})|.$$

The contribution of this atom to the objective gives

$$\begin{aligned}\alpha\phi(t_\omega) &\le \alpha\Phi_\phi(\bar{\mu}_p) && \text{by nonnegativity of the other terms}\\ &\le J(\bar{\mu}) = \alpha\phi\big(T(\bar{\mu})\big).\end{aligned}$$

Strict monotonicity of ϕ gives $t_\omega \le T(\bar{\mu})$. The atom condition in Theorem 3.17 and monotonicity of ϕ' now give

$$|\bar{P}_p(\omega)| = \alpha\phi'(t_\omega) \ge \alpha\phi'\big(T(\bar{\mu})\big).$$

Thus every atom lies in $\mathcal{K}(\bar{\mu})$. On the continuous part, Theorem 3.17 gives

$$|\bar{P}_p(\omega)| = \alpha L_\phi \ge \alpha\phi'\big(T(\bar{\mu})\big) \qquad |(\bar{\mu}_p)_{\mathrm{cont}}|\text{-almost everywhere.}$$

Hence $(\bar{\mu}_p)_{\mathrm{cont}}$ is concentrated on the closed set $\mathcal{K}(\bar{\mu})$.

By (17) and $w_p > 0$, both components of $\bar{\mu}$ are concentrated on the closed set $\mathcal{K}(\bar{\mu})$, which proves (24). □

Theorem 3.20 (Finite support and separation of local minimizers)**.** *Let ϕ satisfy Assumptions 3.2 and 3.3, suppose that $\phi \in C^2((0,\infty))$ satisfies Assumption 3.4(2), let $p > s_1$, and suppose that Assumption 3.1 holds. Then every $\mathcal{M}_p(\Omega)$-local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finite atomic. If $J(\bar{\mu}) > 0$, set*

$$T := T(\bar{\mu}), \qquad \mathcal{K} := \mathcal{K}(\bar{\mu}).$$

Then $\bar{\mu}_{\mathrm{cont}} = 0$, every atom belongs to the compact set $\mathcal{K}$, and any two distinct atoms ω_i and ω_j satisfy

$$\|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} \geq 2\alpha\gamma_T. \tag{25}$$

Consequently,

$$\#\,\mathrm{atom}(\bar{\mu}) \leq N_{\mathrm{cov}}\left(K_p(\mathcal{K}), \frac{1}{2}\sqrt{\alpha\gamma_T}\right) < \infty, \tag{26}$$

where the covering number $N_{\mathrm{cov}}(A, r)$ is the least number of balls of radius r needed to cover A.

Proof. The zero objective case is immediate:

$$J(\bar{\mu}) = 0 \quad\Longrightarrow\quad \Phi_\phi(\bar{\mu}_p) = 0 \quad\Longrightarrow\quad \bar{\mu}_p = 0 \quad\Longrightarrow\quad \bar{\mu} = 0,$$

where the second implication follows from $L_\phi > 0$ and $\phi(t) > 0$ for $t > 0$. We may therefore suppose $J(\bar{\mu}) > 0$. Lemma 3.19 gives the compact set $\mathcal{K}$ and already localizes the atomic and continuous supports. Every normalized atom magnitude

$$t_i := |\bar{\mu}_p(\{\omega_i\})|$$

satisfies $t_i \leq T$ by Lemma 3.19. Proposition 3.7(2), applied with $A = T$, gives

$$L_\phi t - \phi(t) \geq \frac{\gamma_T}{2} t^2 \qquad (0 \leq t \leq T). \tag{27}$$

We first separate distinct atoms. For two such atoms, Theorem 3.13 defines a unique measure $\mu_s \in \mathcal{M}_p(\Omega)$ by

$$(\mu_s)_p := \bar{\mu}_p + s\delta_{\omega_i} - s\delta_{\omega_j}.$$

For sufficiently small $|s|$, the signs of both atom coefficients remain fixed, and

$$\begin{aligned} \|\mu_s - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} &= 2|s|, && \text{by Theorem 3.13,} \\ \mathcal{N}\mu_s &= \mathcal{N}\bar{\mu} + s\big[K_p(\omega_i) - K_p(\omega_j)\big], && \text{by the normalized representation of } \mathcal{N}. \end{aligned}$$

Thus $s \mapsto J(\mu_s)$ is twice differentiable near zero and has a local minimum there. Its second derivative satisfies

$$\begin{aligned} 0 &\leq \frac{d^2}{ds^2} J(\mu_s)\Big|_{s=0} \\ &= \|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} + \alpha\phi''(t_i) + \alpha\phi''(t_j) && \text{by the quadratic fidelity} \\ &\leq \|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} - 2\alpha\gamma_T, && \text{since } t_i, t_j \leq T. \end{aligned}$$

This proves (25).

We next rule out the continuous part by concentrating a local piece, following Pieper and Petrosyan [21, Appendix C]. Suppose $(\bar{\mu}_p)_{\mathrm{cont}} \neq 0$ and write

$$d(\bar{\mu}_p)_{\mathrm{cont}} = \sigma\, d\lambda, \qquad \lambda := |(\bar{\mu}_p)_{\mathrm{cont}}|,$$

where $\sigma \in \{-1, 1\}$ λ-almost everywhere. Without loss of generality, assume $\lambda(\{\sigma = 1\}) > 0$, and restrict λ to this set.

By Theorem 3.17, $\bar{P}_p = -\alpha L_\phi$ holds λ-almost everywhere. Since λ is nonatomic, the countable set of existing atoms has zero λ-measure. We can therefore choose

$$\bar{\omega} \in \operatorname{supp}\lambda \setminus \operatorname{atom}\bar{\mu} \qquad \text{such that} \qquad \bar{P}_p(\bar{\omega}) = -\alpha L_\phi.$$

For $\delta > 0$, put

$$r_\delta := \lambda(B_\delta(\bar{\omega})), \qquad \varepsilon_\delta := \sup_{\omega \in B_\delta(\bar{\omega})} \|K_p(\omega) - K_p(\bar{\omega})\|_{H^1(D)}.$$

Because $\bar{\omega} \in \operatorname{supp}\lambda$, one has $r_\delta > 0$. Moreover, $r_\delta \to \lambda(\{\bar{\omega}\}) = 0$ by continuity from above and the nonatomicity of λ, while $\varepsilon_\delta \to 0$ by continuity of K_p at $\bar{\omega}$. Define μ_δ through its normalized coordinate by

$$(\mu_\delta)_p := \bar{\mu}_p - \lambda|_{B_\delta(\bar{\omega})} + r_\delta \delta_{\bar{\omega}}.$$

The target point is not an atom, and Theorem 3.13 gives

$$\|\mu_\delta - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} = 2r_\delta \longrightarrow 0.$$

If $e_\delta := \mathcal{N}(\mu_\delta - \bar{\mu})$, then

$$\|e_\delta\|_{H^1(D)} \le \varepsilon_\delta r_\delta, \qquad \text{by the Cauchy–Schwarz inequality,}$$

$$\langle r_{\bar{\mu}}, e_\delta \rangle_{H^1(D)} = r_\delta \bar{P}_p(\bar{\omega}) - \int_{B_\delta(\bar{\omega})} \bar{P}_p \, d\lambda = 0, \qquad \text{by Theorem 3.17.}$$

The exact quadratic expansion of the fidelity and the definition of Φ_ϕ therefore give, for all sufficiently small δ such that $r_\delta \le T$,

$$\begin{aligned} J(\mu_\delta) - J(\bar{\mu}) &= \frac{1}{2}\|e_\delta\|^2_{H^1(D)} + \alpha\big[\phi(r_\delta) - L_\phi r_\delta\big] \\ &\le \frac{1}{2}\big(\varepsilon_\delta^2 - \alpha\gamma_T\big) r_\delta^2 && \text{by (27)} \\ &< 0, && \text{for sufficiently small } \delta. \end{aligned}$$

This contradicts local minimality. Hence $\bar{\mu}_{\mathrm{cont}} = 0$.

It remains to count the atoms. The set $K_p(\mathcal{K})$ is compact because $\mathcal{K}$ is compact and K_p is continuous. Cover it by balls of radius $\frac{1}{2}\sqrt{\alpha\gamma_T}$. Each such ball has diameter $\sqrt{\alpha\gamma_T}$, which is strictly smaller than the separation $\sqrt{2\alpha\gamma_T}$ in (25). Hence each ball contains the image of at most one atom. A compact set has a finite cover at every positive radius, and (26) follows. □

Corollary 3.21. *Assume the hypotheses of Theorems 3.16 and 3.20. Then J admits a global minimizer, every global minimizer is finite atomic, and*

$$\min_{\mu \in \mathcal{M}_p(\Omega)} J(\mu) = \min_{\substack{N \ge 0,\ \omega_1,\dots,\omega_N \in \Omega \\ c_1,\dots,c_N \in \mathbb{R}}} \left\{ \frac{1}{2} \left\| \sum_{n=1}^N c_n K(\omega_n) - V \right\|^2_{H^1(D)} + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big) \right\}. \tag{28}$$

The minimum on the right is attained. If μ_* is any global minimizer with $J(\mu_*) > 0$, then it satisfies the following bound

$$\#\operatorname{atom}(\mu_*) \le N_{\mathrm{cov}}\left(K_p\big(\mathcal{K}(\mu_*)\big), \frac{1}{2}\sqrt{\alpha\gamma_{T(\mu_*)}} \right). \tag{29}$$

If $V \neq 0$, define

$$T_0 := \phi^{-1}\left(\frac{\|V\|^2_{H^1(D)}}{2\alpha} \right)$$

and

$$\mathcal{K}_0 := \big\{ \omega \in \Omega : \|V\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} \ge \alpha\phi'(T_0) \big\}. \tag{30}$$

Then $\mathcal{K}_0$ is compact, every atom of every global minimizer lies in $\mathcal{K}_0$, and the following a priori bound holds:

$$\#\operatorname{atom}(\mu_*) \le N_{\mathrm{cov}}\left(K_p(\mathcal{K}_0), \frac{1}{2}\sqrt{\alpha\gamma_{T_0}}\right) < \infty. \tag{31}$$

If $V = 0$, then the zero measure is the unique global minimizer.

Proof. Suppose $V \neq 0$. Then a global minimizer μ_* has positive objective, because Theorem 3.20 would otherwise give $\mu_* = 0$, whose fidelity is positive. By comparison with the zero measure,

$$\begin{aligned} J(\mu_*) \le J(0) &= \frac{1}{2}\|V\|^2_{H^1(D)} = \alpha\phi(T_0), \\ \|r_{\mu_*}\|_{H^1(D)} &\le \|V\|_{H^1(D)}, && \text{since } L(\mu_*) \le J(\mu_*). \end{aligned}$$

Hence

$$T(\mu_*) \le T_0, \qquad \gamma_{T(\mu_*)} \ge \gamma_{T_0}. \tag{32}$$

For an atom ω_i of normalized magnitude $t_i = |\mu_{*,p}(\{\omega_i\})|$, Lemma 3.19 gives $t_i \le T(\mu_*)$. Theorem 3.17 then gives

$$\begin{aligned} \alpha\phi'(T_0) \le \alpha\phi'(t_i) &= |\langle r_{\mu_*}, K_p(\omega_i)\rangle_{H^1(D)}| && \text{since } t_i \le T(\mu_*) \le T_0 \\ &\le \|r_{\mu_*}\|_{H^1(D)}\|K_p(\omega_i)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\le \|V\|_{H^1(D)}\|K_p(\omega_i)\|_{H^1(D)}. \end{aligned}$$

Thus every atom belongs to $\mathcal{K}_0$. The level in (30) is strictly positive, while $K_p \in C_0(\Omega; H^1(D))$, so $\mathcal{K}_0$ is compact. Combining (32) with (25) gives pairwise separation at least $\sqrt{2\alpha\gamma_{T_0}}$. The covering argument from Theorem 3.20 on $K_p(\mathcal{K}_0)$ proves (31).

If $V = 0$, then $J(0) = 0$. Nonnegativity makes zero the minimum, and the conclusion of Theorem 3.20 for $J(\bar{\mu}) = 0$ shows that no other global minimizer exists. □

The hypotheses used so far are jointly satisfiable, and the following family makes the constants explicit.

Example 3.22 (Degeneration of the separation bound at the convex endpoint). For $\gamma \ge 0$, define

$$\phi_\gamma(z) := \begin{cases} \dfrac{z}{2} + \dfrac{1}{4\gamma}\log(1 + 2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases} \tag{33}$$

Then $\phi_\gamma(0) = 0$ and, for $\gamma > 0$,

$$\phi'_\gamma(z) = \frac{1}{2} + \frac{1}{2(1 + 2\gamma z)} \in \left(\tfrac{1}{2}, 1\right], \qquad \phi''_\gamma(t) = -\frac{\gamma}{(1 + 2\gamma t)^2}.$$

Hence $L_{\phi_\gamma} = \phi'_\gamma(0) = 1 \in (0, \infty)$, which is Assumption 3.2; the derivative is positive and nonincreasing and $\phi_\gamma(z) \ge z/2 \to \infty$, which is Assumption 3.3. For Assumption 3.4, the second derivative is continuous and negative on $(0, \infty)$, part (1) holds with $\gamma_1 = \gamma$ and any $\hat{z}_1 > 0$, and part (2) holds with

$$\gamma_A = \frac{\gamma}{(1 + 2\gamma A)^2} > 0. \tag{34}$$

Thus ϕ_γ with $\gamma > 0$ satisfies Assumptions 3.2–3.4, and (34) turns the atom separation $\sqrt{2\alpha\gamma_{T_0}}$ and the neuron bound (31) into computable quantities. This is the penalty used by Algorithm 1.

The limit $\gamma \to 0$ is the convex endpoint. Assumption 3.4 fails at $\gamma = 0$, so Theorem 3.20 and Corollary 3.21 do not apply; the weaker conclusions of Theorems 3.16, 3.17, and 5.1 remain available. The degeneration is quantitative rather than abrupt: by (34), $0 < \gamma_{T_0} \le \gamma \to 0$ as $\gamma \to 0$. Thus the guaranteed separation scale tends to zero; the corresponding covering-number bound may deteriorate or become uninformative.

For fixed network locations $\boldsymbol{\omega} = (\omega_n)_{n=1}^N \in \Omega^N$, define the reduced coefficient objective

$$J_{\boldsymbol{\omega}}(c) := L\left(\sum_{n=1}^N c_n \delta_{\omega_n}\right) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big), \qquad c \in \mathbb{R}^N. \tag{35}$$

Theorem 3.23 (Sufficient condition for local optimality)**.** *Let $\phi : [0,\infty) \to [0,\infty)$ satisfy $\phi(0) = 0$ and $\phi \in C^1((0,\infty))$, with $0 < L_\phi := \phi'(0+) < \infty$. Suppose there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that*

$$\phi(z) \geq L_\phi z - \frac{\gamma_1}{2} z^2 \qquad (0 \leq z \leq \hat{z}_1). \tag{36}$$

Let $p > s_1$, let $V \in H^1(D)$, and suppose that Assumption 3.1 holds. Let

$$\bar{\boldsymbol{\omega}} := (\bar{\omega}_n)_{n=1}^N, \qquad \bar{c} := (\bar{c}_n)_{n=1}^N \in \mathbb{R}^N, \qquad \bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n}.$$

Suppose also that

$$\big|\phi'\big(w_p(\bar{\omega}_n)|\bar{c}_n|\big)\big| < L_\phi \qquad (n = 1, \dots, N). \tag{37}$$

Assume that

(1) *$\bar{c}$ is a local minimizer of $J_{\bar{\boldsymbol{\omega}}}$;*
(2) *the strict inequality*

$$|\bar{P}_p(\omega)| < \alpha L_\phi$$

holds for every $\omega \in \Omega \setminus \operatorname{atom} \bar{\mu}$.

Then $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$-local minimizer of J.

Proof. We adapt the proof of Pieper and Petrosyan [21, Theorem 4] in the normalized coordinate of Theorem 3.13. Boundedness of K_p in $H^1(D)$ allows their quadratic loss estimates with $H^1(D)$ in place of L^2. Stationarity in condition (1), (37), and condition (2) give $|\bar{P}_p| < \alpha L_\phi$ everywhere. Since $\bar{P}_p \in C_0(\Omega)$, the gap below αL_ϕ is uniform; this replaces compactness of the parameter domain in their proof. The bound (36) supplies their penalty estimate for small perturbations, with L_ϕ in place of 1. Their remaining argument gives local minimality of $\bar{\mu}_p$ with respect to the total variation norm, and the normalization isometry yields the claim. □

Admissible activations. The two exponents in Assumption 3.1 play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is used by Theorems 3.17, 3.20, and 5.1. Since $\|K(\omega)\|_{L^2(D)} \leq \|K(\omega)\|_{H^1(D)}$, it is consistent to impose $s_0 \leq s_1$, as Assumption 3.1 does. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ normalizes the full H^1 atom; together with Assumption 3.4(2), it is sufficient for the conclusion of Theorem 3.20 that local minimizers have finite support. The following exponents satisfy the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$\mathrm{GELU}(z)^2$	2	2	$p > 2$

For bounded activations such as tanh and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate.

4. The k-homogeneous formulation

The ReLU^k activation function is of particular interest since it is widely used in Sobolev training [15, 12]. For gradient fitting with ReLU^k we use $k \geq 2$, so that $K(\omega) \in C^1(\overline{D})$. As we will see in this section, the induced penalty is $\psi_k(t) = t^q$ with $q = 2/(k+1) < 1$, so it has infinite right slope at zero and does not extend the result in Section 3.

On the other hand, k-homogeneity makes the radial part of an inner parameter redundant: every atom can be moved onto the unit sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ at the cost of rescaling its outer coefficient. The parameter domain of the resulting problem is compact, so the dictionary is bounded, and neither the polynomial weight w_p nor the weighted space $\mathcal{M}_p(\Omega)$ of Section 3 is required. Throughout this section we work on $\mathcal{M}(\mathbb{S}^d)$ with the total variation norm $\|\cdot\|_{\mathcal{M}(\mathbb{S}^d)}$, which we identify with the subspace of $\mathcal{M}(\Omega)$ of measures vanishing outside $\mathbb{S}^d$.

4.1. The measure problem on the sphere.

Assumption 4.1 (k-homogeneity)**.** For some $k \geq 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \qquad (\lambda > 0,\ \omega \in \mathbb{R}^{d+1}).$$

Lemma 4.2 (The ReLU^k dictionary)**.** *Let $k \geq 2$, let $D \subset \mathbb{R}^d$ be a bounded domain, let $\rho(z) = (z_+)^k$, and let*

$$[K(\omega)](x) = (a \cdot x + b)_+^k, \qquad \omega = (a, b) \in \mathbb{R}^{d+1}.$$

Then

1. *K satisfies Assumption 4.1 with degree k;*
2. *$K(\omega) \in C^1(\mathbb{R}^d)$ with $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a$, and this classical gradient is also the distributional gradient of $K(\omega)$ on D.*

Proof. (1) This follows from $(\lambda z)_+ = \lambda z_+$ for $\lambda > 0$.

(2) Since $k \geq 2$, $\rho \in C^1(\mathbb{R})$ with $\rho'(z) = k(z_+)^{k-1}$, so the chain rule gives $K(\omega) \in C^1(\mathbb{R}^d)$ and the stated gradient formula. Both $K(\omega)$ and its gradient are bounded on D and hence square-integrable; integration by parts identifies the classical gradient with the weak gradient. □

Theorem 4.3 (Boundedness and continuity of the value operator)**.** *The* ReLU^k *dictionary of Lemma 4.2 is continuous from $\mathbb{S}^d$ into $H^1(D)$. The constant*

$$C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}$$

is finite and strictly positive. The value operator $\mathcal{N} : \mathcal{M}(\mathbb{S}^d) \to H^1(D)$ is well defined, bounded, and linear, with

$$\|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} \qquad \big(\mu \in \mathcal{M}(\mathbb{S}^d)\big). \tag{38}$$

Proof. By Lemma 4.2(2), the atoms and their gradients are jointly continuous on $\mathbb{S}^d \times \overline{D}$. Uniform continuity on this compact set and $|D| < \infty$ give continuity of $K : \mathbb{S}^d \to H^1(D)$. Compactness of $\mathbb{S}^d$ implies $C_S < \infty$, and $K(0, 1) \equiv 1$ gives $C_S \geq |D|^{1/2} > 0$.

Since K is continuous and bounded, the Bochner integral $\mathcal{N}\mu$ exists for every $\mu \in \mathcal{M}(\mathbb{S}^d)$. Linearity and the integral norm estimate give (38). □

The following lemma extends the degree-one ReLU rescaling result [21, Appendix F, Proposition 3] to a k-homogeneous dictionary. We retain general positive exponents to make explicit how the homogeneity degree changes both the induced exponent and its coefficient.

Lemma 4.4 (Rescaling of a k-homogeneous network)**.** *Let Assumption 4.1 hold, and let $\xi, \zeta > 0$. For a fixed nonzero represented coefficient $\widetilde{c}$ and a unit direction $\widehat{\omega} \in \mathbb{S}^d$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda\widehat{\omega}, \qquad \lambda > 0, \qquad c = \widetilde{c}\lambda^{-k}.$$

The minimum equals

$$C_{k,\xi,\zeta}|\widetilde{c}|^q, \qquad q := \frac{\xi\zeta}{k\xi+\zeta}, \qquad C_{k,\xi,\zeta} := \frac{1}{q}\,k^{-k\xi/(k\xi+\zeta)}. \tag{39}$$

Consequently, consider any finite representation with nonzero inner parameters and set

$$\widehat{\omega}_n := \frac{\omega_n}{|\omega_n|}, \qquad \widetilde{c}_n := c_n|\omega_n|^k.$$

Among all independent radial rescalings of this form, the minimum total penalty is $C_{k,\xi,\zeta}\sum_n |\widetilde{c}_n|^q$.

Proof. We follow the rescaling argument of [21, Appendix F, proof of Proposition 3]. The radial penalty and its derivative are

$$g(\lambda) = \frac{|\widetilde{c}|^\xi}{\xi\lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta},$$
$$g'(\lambda) = \lambda^{-k\xi-1}\left(\lambda^{k\xi+\zeta} - k|\widetilde{c}|^\xi\right).$$

Since $\lambda^{-k\xi-1} > 0$ for $\lambda > 0$ and the bracketed term is strictly increasing from $-k|\widetilde{c}|^\xi < 0$ to $+\infty$, g' changes sign exactly once, from negative to positive, at

$$\lambda_* = \left(k|\widetilde{c}|^\xi\right)^{1/(k\xi+\zeta)}.$$

Thus g has its unique global minimum at λ_*, and substitution gives $g(\lambda_*) = C_{k,\xi,\zeta}|\widetilde{c}|^q$. The radial variables are independent, so summing their minima proves the assertion for finite networks. □

Remark 4.5 (The quadratic case and the meaning of α). For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \qquad C_k := C_{k,2,2} = \frac{k+1}{2}\,k^{-k/(k+1)}.$$

In particular $C_1 = 1$ and $C_2 = \frac{3}{2}\,2^{-2/3}$. Since C_k depends on k, equal nominal values of α in the penalty $\alpha\sum_n |c_n|^q$ for different powers k do *not* correspond to equal coefficients of the original separable penalty.

In the remainder of this section, K denotes the ReLU^k dictionary of Lemma 4.2. For $k \geq 2$ define the scalar penalty

$$\psi_k(t) := t^q, \qquad q := \frac{2}{k+1} \in (0,1), \qquad t \geq 0.$$

Here q is the exponent obtained in Lemma 4.4 when $\xi = \zeta = 2$. It is continuous, strictly increasing, and concave, with $\psi_k(0) = 0$ and $\psi_k'(0^+) = +\infty$. By Lemma 3.6(1),

$$\psi_k(s+t) \leq \psi_k(s) + \psi_k(t) \qquad (s,t \geq 0). \tag{40}$$

Iterating (40) and using continuity gives, for every summable sequence $(t_j)_{j\geq 1}$ of nonnegative numbers,

$$\psi_k\Big(\sum_{j\geq 1} t_j\Big) \leq \sum_{j\geq 1} \psi_k(t_j). \tag{41}$$

Definition 4.6 (Homogeneous measure penalty)**.** For $\mu \in \mathcal{M}(\mathbb{S}^d)$ with atomic and continuous parts μ_{atom} and μ_{cont} as in Section 2.1, the atom set is at most countable. Hence the following sum of nonnegative terms is well defined in $[0,+\infty]$ and independent of its enumeration. Set

$$\Phi_{\psi_k}(\mu) := \begin{cases} \displaystyle\sum_{\omega\in\mathrm{atom}\,\mu} \psi_k\big(|\mu(\{\omega\})|\big), & \mu_{\mathrm{cont}} = 0, \\ +\infty, & \mu_{\mathrm{cont}} \neq 0. \end{cases}$$

The functional is nonnegative, vanishes only at the zero measure, and, whenever $\Phi_{\psi_k}(\mu) < \infty$, countable subadditivity gives

$$\Phi_{\psi_k}(\mu) \geq \psi_k\big(\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}\big) = \|\mu\|^q_{\mathcal{M}(\mathbb{S}^d)}. \tag{42}$$

For $\mu \in \mathcal{M}(\mathbb{S}^d)$ we now set

$$J_{\psi_k}(\mu) := L(\mu) + \alpha\Phi_{\psi_k}(\mu).$$

By Definition 4.6, every measure with finite objective is purely atomic, possibly with countably infinitely many atoms. Since the empirical fidelity is also one half of a squared norm in a Hilbert space and its dictionary is continuous on the compact sphere, all results below hold with L replaced by l^M and the corresponding sampled norms and bounds. For ReLU^k with $k \geq 2$, the sampled atom values and gradients are continuous and uniformly bounded on $\mathbb{S}^d$ by Lemma 4.2(2) and compactness. Thus the empirical fidelity (7) extends to all $\mu \in \mathcal{M}(\mathbb{S}^d)$ by integrating these sampled quantities against μ.

We study the minimization problem

$$\min_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu). \tag{P_{ψ_k}}$$

For a finite network on $\mathbb{S}^d$, the definition of Φ_{ψ_k} gives

$$J_{\psi_k}(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} |c_n|^q.$$

4.2. **Global existence and a uniform width bound.** Local minimizers of (P_{ψ_k}) are understood as in Definition 3.14, applied to J_{ψ_k} on $\mathcal{M}(\mathbb{S}^d)$, equipped with the total variation norm.

Lemma 4.7. *For $N \in \mathbb{N}_0$, $\boldsymbol{\omega} \in (\mathbb{S}^d)^N$, and $c \in \mathbb{R}^N$, define the finitely supported measure $u^N_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Then the infima over all measures, purely atomic measures, and finitely supported measures coincide:*

$$\inf_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu) = \inf_{\substack{\mu \in \mathcal{M}(\mathbb{S}^d) \\ \mu_{\mathrm{cont}}=0}} J_{\psi_k}(\mu) = \inf_{\substack{N \in \mathbb{N}_0 \\ \boldsymbol{\omega} \in (\mathbb{S}^d)^N,\ c \in \mathbb{R}^N}} J_{\psi_k}(u^N_{\boldsymbol{\omega},c}). \tag{43}$$

Proof. Definition 4.6 allows the infimum over all measures to be restricted to purely atomic measures.

Every finitely supported measure is purely atomic, so

$$\inf_{\substack{\mu \in \mathcal{M}(\mathbb{S}^d) \\ \mu_{\mathrm{cont}}=0}} J_{\psi_k}(\mu) \leq \inf_{\substack{N \in \mathbb{N}_0 \\ \boldsymbol{\omega} \in (\mathbb{S}^d)^N,\ c \in \mathbb{R}^N}} J_{\psi_k}(u^N_{\boldsymbol{\omega},c}).$$

For the converse, let $J_{\psi_k}(\mu) < \infty$ and write its atomic representation as $\mu = \sum_{j \in \mathcal{I}} c_j \delta_{\omega_j}$, $\mathcal{I} \subseteq \mathbb{N}$. The truncations $\mu_m := \sum_{j \in \mathcal{I},\, j \leq m} c_j \delta_{\omega_j}$ satisfy $\mu_m \to \mu$ in total variation and $\Phi_{\psi_k}(\mu_m) \to \Phi_{\psi_k}(\mu)$ by convergence of the total variation and penalty series. The bound (38) then gives $L(\mu_m) \to L(\mu)$. Each μ_m is finitely supported, so the infimum over finitely supported measures is at most $J_{\psi_k}(\mu_m)$ for every m. Passing to the limit yields

$$\inf_{\substack{N \in \mathbb{N}_0 \\ \boldsymbol{\omega} \in (\mathbb{S}^d)^N,\ c \in \mathbb{R}^N}} J_{\psi_k}(u^N_{\boldsymbol{\omega},c}) \leq \lim_{m \to \infty} J_{\psi_k}(\mu_m) = J_{\psi_k}(\mu).$$

Taking the infimum over all μ with finite objective proves the reverse inequality. □

Theorem 4.8 (Stationarity and a lower bound on atom weights)**.** *Let $k \geq 2$, $q = 2/(k+1) \in (0,1)$, $\alpha > 0$, and $V \in H^1(D)$. Let $\bar{\mu} \in \mathcal{M}(\mathbb{S}^d)$ satisfy $J_{\psi_k}(\bar{\mu}) < \infty$ and be optimal against perturbations of the weights of its own atoms: there is $\varepsilon > 0$ such that*

$$J_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu} + t\delta_\omega) \qquad \text{for every } \omega \in \operatorname{atom} \bar{\mu} \text{ and every } |t| < \varepsilon. \tag{44}$$

Set

$$B(\bar{\mu}) := \|P_{\bar{\mu}}\|_{C(\mathbb{S}^d)}.$$

Then $B(\bar{\mu}) < \infty$, *and every atom* $\omega_n \in \operatorname{atom}\bar{\mu}$ *with weight* $c_n := \bar{\mu}(\{\omega_n\})$ *satisfies*

$$P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1} \operatorname{sign}(c_n) = 0. \tag{45}$$

If $B(\bar{\mu}) = 0$, *then* $\bar{\mu} = 0$. *If* $B(\bar{\mu}) > 0$, *then*

$$|c_n| \geq c_{\min}(\bar{\mu}) := \left(\frac{\alpha q}{B(\bar{\mu})}\right)^{1/(1-q)} > 0 \qquad \text{for every } \omega_n \in \operatorname{atom}\bar{\mu}. \tag{46}$$

Hypothesis (44) *holds in particular if* $\bar{\mu}$ *is a local minimizer of* (P_{ψ_k}).

Proof. The map $P_{\bar{\mu}}$ is continuous on $\mathbb{S}^d$, being the composition of the continuous map K with the continuous linear functional $\langle r_{\bar{\mu}}, \cdot\rangle_{H^1(D)}$, and $\mathbb{S}^d$ is compact, so $B(\bar{\mu}) < \infty$.

Fix $\omega_n \in \operatorname{atom}\bar{\mu}$ and $s \in \{-1, 1\}$, and consider $\mu_t := \bar{\mu} + ts\delta_{\omega_n}$ for $0 < t < \min\{\varepsilon, |c_n|\}$,

$$\Phi_{\psi_k}(\mu_t) - \Phi_{\psi_k}(\bar{\mu}) = \psi_k(|c_n + ts|) - \psi_k(|c_n|).$$

For the fidelity,

$$\begin{aligned} L(\mu_t) - L(\bar{\mu}) &= ts\langle r_{\bar{\mu}}, K(\omega_n)\rangle_{H^1(D)} + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)} \\ &= tsP_{\bar{\mu}}(\omega_n) + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)}. \end{aligned}$$

Because $c_n \neq 0$, the map $z \mapsto |z|^q$ is differentiable at c_n with derivative $q|c_n|^{q-1}\operatorname{sign}(c_n)$. Hypothesis (44) gives $J_{\psi_k}(\mu_t) - J_{\psi_k}(\bar{\mu}) \geq 0$; dividing by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s\Big[P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1}\operatorname{sign}(c_n)\Big].$$

Choosing $s = 1$ and $s = -1$ gives (45).

Suppose $B(\bar{\mu}) = 0$, that is, $P_{\bar{\mu}} \equiv 0$ on $\mathbb{S}^d$. If $\bar{\mu}$ had an atom ω_n, then (45) would read $\alpha q|c_n|^{q-1}\operatorname{sign}(c_n) = 0$, which is impossible because $\alpha > 0$, $q > 0$, and $|c_n|^{q-1} \in (0, \infty)$ for $c_n \neq 0$. Hence $\operatorname{atom}\bar{\mu} = \emptyset$, and since $\bar{\mu}$ is purely atomic, $\bar{\mu} = 0$.

Suppose $B(\bar{\mu}) > 0$. Taking absolute values in (45) gives

$$\alpha q |c_n|^{q-1} = |P_{\bar{\mu}}(\omega_n)| \leq B(\bar{\mu}).$$

Since $q - 1 = -(1 - q) < 0$, this reads $|c_n|^{-(1-q)} \leq B(\bar{\mu})/(\alpha q)$, hence $|c_n|^{1-q} \geq \alpha q / B(\bar{\mu})$; raising both sides to the positive power $1/(1-q)$ preserves the inequality and gives (46). □

Remark 4.9. For measures with finite objective, the infinite right slope $\psi_k'(0^+) = +\infty$ has the following consequences for atom insertion.

(1) Inserting a new atom at $\omega \notin \operatorname{atom}\bar{\mu}$ with weight ts, $s \in \{-1, 1\}$, changes the penalty by αt^q. Since $t^q/t = t^{q-1} \to \infty$ as $t \downarrow 0$, division by t produces no finite limit, and hence no pointwise condition on $P_{\bar{\mu}}(\omega)$ away from $\operatorname{atom}\bar{\mu}$. Theorem 4.8 is restricted to existing atoms for this reason.

(2) The zero measure is a local minimizer of (P_{ψ_k}) in total variation. Indeed, $J_{\psi_k}(\mu) = \infty$ whenever $\Phi_{\psi_k}(\mu) = \infty$, while for μ with $\Phi_{\psi_k}(\mu) < \infty$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \varepsilon$,

$$\begin{aligned} J_{\psi_k}(\mu) - J_{\psi_k}(0) &= \langle -V, \mathcal{N}\mu\rangle_{H^1(D)} + \tfrac{1}{2}\|\mathcal{N}\mu\|^2_{H^1(D)} + \alpha\Phi_{\psi_k}(\mu) \\ &\geq -\|V\|_{H^1(D)} C_S\, \varepsilon + \alpha\varepsilon^q, \end{aligned}$$

by (42), (38), and the Cauchy–Schwarz inequality. Since $q < 1$, the right-hand side is positive for every sufficiently small $\varepsilon > 0$.

An insertion criterion based on an infinitesimal weight perturbation is therefore unavailable for this penalty.

Theorem 4.10 (Existence and width bound)**.** *Let $k \geq 2$, $q = 2/(k+1) \in (0,1)$, $\alpha > 0$, let $V \in H^1(D)$ with $V \neq 0$. Then (P_{ψ_k}) admits a global minimizer, and its minimum value equals the infimum over finite networks on the right-hand side of (43). Every global minimizer $\bar{\mu}$ is a finite atomic measure satisfying*

$$\#\operatorname{atom}\bar{\mu} \leq N_{\max} := \frac{\|V\|_{H^1(D)}^2}{2\alpha\, c_{\min,\mathrm{glob}}^q}, \qquad c_{\min,\mathrm{glob}} := \left(\frac{\alpha q}{\|V\|_{H^1(D)}\, C_S}\right)^{1/(1-q)}.$$

Remark 4.11. If $V = 0$, then $\mu = 0$ is the unique global minimizer by the nonnegativity of the fidelity and the strict positivity of Φ_{ψ_k} away from zero. Thus the atom bound is zero in this case. The assumption $V \neq 0$ is needed only to ensure that $c_{\min,\mathrm{glob}} \in (0,\infty)$ and $N_{\max} < \infty$ are well defined.

Proof. For $N \in \mathbb{N}_0$, set

$$F_N(\boldsymbol{\omega}, c) := L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} |c_n|^q, \qquad m_N := \inf F_N,$$

with $F_0 = m_0 = \frac{1}{2}\|V\|_{H^1(D)}^2$.

For $N \geq 1$, continuity of K, scalar multiplication, and finite addition show that $(\boldsymbol{\omega}, c) \mapsto \sum_{n=1}^{N} c_n K(\omega_n)$ is continuous into $H^1(D)$; hence F_N is continuous.

A minimizing sequence for F_N may be chosen in the sublevel $F_N \leq m_0 + 1$. Since the fidelity is nonnegative, this sublevel satisfies

$$|c_n| \leq R := \left(\frac{m_0 + 1}{\alpha}\right)^{1/q} \qquad (n = 1, \dots, N).$$

It therefore lies in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. Compactness gives a convergent subsequence, and continuity shows that its limit attains m_N. The case $N = 0$ is immediate.

Suppose that a measure μ satisfies $J_{\psi_k}(\mu) \leq m_0$ and the atomwise optimality hypothesis (44). The nonnegativity of the penalty gives

$$\frac{1}{2}\|r_\mu\|_{H^1(D)}^2 = L(\mu) \leq J_{\psi_k}(\mu) \leq m_0 = \frac{1}{2}\|V\|_{H^1(D)}^2.$$

Applying (38) to Dirac masses and using Cauchy–Schwarz yields

$$B(\mu) = \sup_{\omega \in \mathbb{S}^d} \left|\langle r_\mu, K(\omega)\rangle_{H^1(D)}\right| \leq \|r_\mu\|_{H^1(D)} C_S \leq \|V\|_{H^1(D)} C_S.$$

If $\mu \neq 0$, Theorem 4.8 gives $B(\mu) > 0$ and

$$|\mu(\{\omega\})| \geq \left(\frac{\alpha q}{B(\mu)}\right)^{1/(1-q)} \geq c_{\min,\mathrm{glob}} \qquad (\omega \in \operatorname{atom}\mu).$$

Consequently, for every finite $A \subseteq \operatorname{atom}\mu$,

$$\#A\, \alpha c_{\min,\mathrm{glob}}^q \leq \alpha \sum_{\omega \in A} |\mu(\{\omega\})|^q \leq \alpha \Phi_{\psi_k}(\mu) \leq J_{\psi_k}(\mu) \leq \frac{1}{2}\|V\|_{H^1(D)}^2.$$

An infinite atom set would contain finite subsets of arbitrarily large cardinality. Thus $\operatorname{atom}\mu$ is finite and its cardinality is at most $N_{\max}$; the same conclusion is immediate when $\mu = 0$.

The bound above applies to every minimizer of F_N: its value is at most m_0, and its coefficients can be varied independently. Each such minimizer therefore has at most $\lfloor N_{\max} \rfloor$ atoms. Hence

$$\inf_{N \in \mathbb{N}_0} m_N = \min_{\substack{N \in \mathbb{N}_0 \\ N \leq \lfloor N_{\max} \rfloor}} m_N.$$

Since each m_N is attained, Lemma 4.7 gives a global minimizer of (P_{ψ_k}). Every global minimizer μ satisfies $J_{\psi_k}(\mu) \leq m_0$ and (44), so the bound derived above gives $\#\operatorname{atom}\mu \leq N_{\max}$. □

Corollary 4.12 (Finite support of local minimizers with respect to the total variation norm)**.** *Let the hypotheses of Theorem 4.8 on k, q, α, V hold and let $\bar{\mu} \neq 0$ be a local minimizer of (P_{ψ_k}) in total variation. Then $B(\bar{\mu}) > 0$, the measure $\bar{\mu}$ is finite atomic, and*

$$\#\operatorname{atom} \bar{\mu} \leq \frac{J_{\psi_k}(\bar{\mu})}{\alpha\, c_{\min}(\bar{\mu})^{\,q}}.$$

Proof. By Theorem 4.8, $B(\bar{\mu}) = 0$ would force $\bar{\mu} = 0$; hence $B(\bar{\mu}) > 0$ and every atom weight satisfies $|c_n| \geq c_{\min}(\bar{\mu}) > 0$. Since ψ_k is nondecreasing and the fidelity is nonnegative,

$$N\alpha\, c_{\min}(\bar{\mu})^{\,q} \leq \alpha \Phi_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu}) < \infty$$

with $N = \#\operatorname{atom} \bar{\mu} \in \mathbb{N}_0 \cup \{\infty\}$, which forces $N < \infty$ and gives the stated bound. □

The bound of Corollary 4.12 depends on the particular minimizer, whereas $N_{\max}$ in Theorem 4.10 is uniform over all global minimizers and depends on the data.

5. Algorithms

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted, and different insertion methods are chosen given the problem setting. In the algorithms and numerical examples, J_{ψ_k} uses the empirical fidelity l^M from (7) in place of L.

Data generation via the open-loop problem. Set $U := L^2(t_0, T; \mathbb{R}^{m_u})$ and assume that for every $u \in U$ the state equation in (P) has a unique solution $y(u)$ on $[t_0, T]$ and that the control-to-state map $u \mapsto y(u)$ is continuously Fréchet differentiable from U into $C([t_0, T]; \mathbb{R}^d)$. The objective of (P) is then the reduced functional $u \mapsto \mathcal{C}(u; t_0, x)$ on U, and standard adjoint (Lagrangian) calculus [13] gives its gradient

$$\nabla_u \mathcal{C}(u; t_0, x) = 2\eta u + g(y(u))^\top p \quad \text{in } U,$$

where p solves the costate equation of (TPBVP) backward in time along $(y(u), u)$ in place of (y^*, u^*). Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the costate equation, and the implicit derivative of $u \mapsto y(u)$ is never formed. [14] surveys generation methods.

The two algorithms. We use two algorithms of primal–dual active point (PDAP) type, in the sense of the accelerated conditional gradient method for measure spaces of [22]. Algorithm 1 follows the nonconvex variant of [21], whose scalar penalty satisfies $\phi'(0) < \infty$ as in Assumption 3.2; Algorithm 2 is an adaptation of it to the penalty ψ_k, whose infinite slope at the origin is explicitly excluded by that reference.

In both algorithms, we hold the locations $\boldsymbol{\omega}$ fixed and optimize the outer weights using a semismooth Newton method. They are specified in their respective subsections below. In either case the method is local and its output is retained only when it does not increase the objective relative to its value immediately after insertion.

For the sampled fidelity, write $P_\mu^M(\omega) := Dl^M(\mu)[\delta_\omega]$, and let $\|K(\omega)\|_M$ denote the empirical H^1 seminorm associated with l^M.

For a joint insertion $\sum_{i=1}^r c_i \delta_{\omega_i}$ at new locations the penalty is additive, but the fidelity change contains pairwise cross terms between the inserted atoms, which no individual test controls. For $N_{\text{ins}} = 1$ the individual decrease statement applies to every accepted candidate, while Corollary 5.2 additionally requires an exact global maximizer.

5.1. **Algorithm 1: the log penalty on the normalized measure.** The insertion step seeks a location ω and a coefficient c such that $J(\mu + c\delta_\omega) < J(\mu)$. Recall from Section 3.5 that

$$P_{p,\mu}(\omega) = \langle r_\mu, K_p(\omega)\rangle_{H^1(D)}, \qquad B_p = \sup_{\omega\in\Omega} \|K_p(\omega)\|_{H^1(D)}.$$

Here $P_{p,\mu}(\omega)$ is the fidelity derivative with respect to the normalized inserted coefficient $\kappa = w_p(\omega)c$. At a local minimizer, the necessary condition (21) requires $|P_{p,\mu}(\omega)| \le \alpha L_\phi$ at every location, where $L_\phi = \phi'(0+)$. We use a violation of this inequality to select a candidate for insertion.

For $p > s_1$ and $\mu \in \mathcal{M}_p(\Omega)$, define

$$\Delta(\mu,\omega) := \max\{|P_{p,\mu}(\omega)| - \alpha L_\phi, 0\},$$
$$\Delta(\mu) := \sup_{\omega\in\Omega} \Delta(\mu,\omega) = \max\{\|P_{p,\mu}\|_\infty - \alpha L_\phi, 0\}.$$

We call $\Delta(\mu)$ the first-order optimality residual for atom insertion; $\Delta(\mu,\omega)$ is the corresponding residual at a single location. Thus $\Delta(\mu) = 0$ means that the necessary insertion inequality holds everywhere, but does not by itself establish local optimality. The following theorem converts a positive residual into an explicit insertion coefficient and a guaranteed objective decrease. It also bounds the search region and proves that $|P_{p,\mu}|$ attains its maximum there whenever $\Delta(\mu) > 0$.

Theorem 5.1 (Decrease and search region for a single-atom insertion)**.** *Suppose that Assumption 3.1 holds, $p > s_1$, and $\alpha > 0$. Let ϕ satisfy Assumptions 3.2 and 3.3, and fix $\mu \in \mathcal{M}_p(\Omega)$.*

(1) *If $\Delta(\mu,\omega) > 0$, then $B_p > 0$, and inserting $c(\omega)\delta_\omega$ with $c(\omega) := -\frac{\Delta(\mu,\omega)}{w_p(\omega)B_p^2}\operatorname{sign}(P_{p,\mu}(\omega))$ decreases J by at least $\frac{\Delta(\mu,\omega)^2}{2B_p^2}$.*

(2) *Set $R(\mu) := \max\{1, (\frac{2^{s_1+1}C_K\|r_\mu\|_{H^1(D)}}{\alpha L_\phi})^{\frac{1}{p-s_1}}\}$. Every ω with $\Delta(\mu,\omega) > 0$ belongs to $\overline{B}_{R(\mu)}$. If $\Delta(\mu) > 0$, there exists $\omega^* \in \overline{B}_{R(\mu)}$ such that*

$$|P_{p,\mu}(\omega^*)| = \|P_{p,\mu}\|_\infty, \qquad \Delta(\mu,\omega^*) = \Delta(\mu).$$

At this location, the insertion in part (1) decreases J by at least $\frac{\Delta(\mu)^2}{2B_p^2}$.

Proof. We first prove the decrease estimate by bounding the penalty increment and expanding the quadratic fidelity. $p > s_1$ and (4) give $K_p \in C_0(\Omega; H^1(D))$, hence $B_p < \infty$ and $P_{p,\mu} \in C_0(\Omega)$. If $B_p = 0$, then $K_p \equiv 0$ and $P_{p,\mu} \equiv 0$. Thus $\Delta(\mu,\omega) > 0$ implies $B_p > 0$.

For $\kappa \in \mathbb{R}$, adding $\kappa\delta_\omega$ changes only the mass at ω. Monotonicity and Lemma 3.6 give

$$\begin{aligned} \Phi_\phi(\mu_p + \kappa\delta_\omega) - \Phi_\phi(\mu_p) &\le \phi(|\kappa|) && \text{by subadditivity at the point } \omega \\ &\le L_\phi|\kappa| && \text{by the tangent bound at zero.} \end{aligned} \tag{47}$$

Since $(\mu + \kappa w_p(\omega)^{-1}\delta_\omega)_p = \mu_p + \kappa\delta_\omega$, the quadratic fidelity has the exact expansion

$$\begin{aligned} &J\left(\mu + \frac{\kappa}{w_p(\omega)}\delta_\omega\right) - J(\mu) \\ &\quad\le \kappa P_{p,\mu}(\omega) + \frac{\kappa^2}{2}\|K_p(\omega)\|_{H^1(D)}^2 + \alpha L_\phi|\kappa| && \text{by (47)} \\ &\quad\le \kappa P_{p,\mu}(\omega) + \frac{B_p^2\kappa^2}{2} + \alpha L_\phi|\kappa| && \text{by the definition of } B_p. \end{aligned} \tag{48}$$

Fix $\omega \in \Omega$ with $\Delta(\mu,\omega) > 0$ and write $\kappa(\omega) := w_p(\omega)c(\omega)$. Then $|P_{p,\mu}(\omega)| = \alpha L_\phi + \Delta(\mu,\omega)$, and substitution of $\kappa = \kappa(\omega)$ from part (1) into (48) yields

$$\begin{aligned} &J\big(\mu + c(\omega)\delta_\omega\big) - J(\mu) \\ &\quad\le -|\kappa(\omega)|\big(\alpha L_\phi + \Delta(\mu,\omega)\big) + \frac{B_p^2|\kappa(\omega)|^2}{2} + \alpha L_\phi|\kappa(\omega)| \quad \text{since } \operatorname{sign}\kappa(\omega) = -\operatorname{sign} P_{p,\mu}(\omega) \\ &\quad= -|\kappa(\omega)|\Delta(\mu,\omega) + \frac{B_p^2|\kappa(\omega)|^2}{2} \end{aligned}$$

$$= -\frac{\Delta(\mu,\omega)^2}{2B_p^2} \qquad \text{since } |\kappa(\omega)| = \frac{\Delta(\mu,\omega)}{B_p^2}.$$

This proves part (1).

We now bound the search region using (4) and establish attainment on the resulting compact ball. If $|\omega| \geq 1$, then Cauchy–Schwarz gives

$$\begin{aligned} |P_{p,\mu}(\omega)| &\leq \|r_\mu\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\leq C_K \|r_\mu\|_{H^1(D)} \frac{(1+|\omega|)^{s_1}}{1+|\omega|^p} && \text{by (4)} \\ &\leq 2^{s_1} C_K \|r_\mu\|_{H^1(D)} |\omega|^{-(p-s_1)} && \text{since } |\omega| \geq 1. \end{aligned}$$

If $|\omega| > R(\mu)$, the preceding estimate and the definition of $R(\mu)$ give $|P_{p,\mu}(\omega)| < \frac{\alpha L_\phi}{2}$. Hence $\Delta(\mu,\omega) = 0$. This proves the claimed localization. If $\Delta(\mu) > 0$, some location has $|P_{p,\mu}(\omega)| > \alpha L_\phi$. Such a location lies in $\overline{B}_{R(\mu)}$, since outside this ball $|P_{p,\mu}| < \frac{\alpha L_\phi}{2}$. The continuous function $|P_{p,\mu}|$ attains a maximum on this compact ball. This maximum exceeds αL_ϕ and hence every value outside the ball, so it is global. Its location ω^* satisfies $\Delta(\mu,\omega^*) = \Delta(\mu)$; applying part (1) there proves the remaining decrease claim. □

Corollary 5.2. *Let $(\mu_k)_{k\geq 0}$ be finite atomic iterates with $J(\mu_0) < \infty$, and write $\Delta_k := \Delta(\mu_k)$. If $\Delta_k > 0$, use Theorem 5.1 at μ_k to select an exact maximizer ω^* and insert $c(\omega^*)\delta_{\omega^*}$. Then apply a monotone coefficient correction, which does not increase the objective relative to the iterate immediately after insertion. Then*

$$\sum_{k=0}^{\infty} \Delta_k^2 \leq 2B_p^2 J(\mu_0),$$

$$\Delta_k \longrightarrow 0,$$

$$\min_{0\leq k<n} \Delta_k \leq B_p \sqrt{\frac{2J(\mu_0)}{n}} \qquad (n \geq 1).$$

These conclusions do not assert convergence of (μ_k) to a global minimizer or an objective gap rate.

Proof. If $B_p = 0$, then $\Delta_k = 0$ for every k and the claims are immediate. Otherwise Theorem 5.1 and the correction hypothesis give, for every $n \geq 1$,

$$\sum_{k=0}^{n-1} \frac{\Delta_k^2}{2B_p^2} \leq J(\mu_0) - J(\mu_n) \leq J(\mu_0).$$

Letting $n \to \infty$ gives the first claim, and summability gives $\Delta_k \to 0$. Finally,

$$\min_{0\leq k<n} \Delta_k^2 \leq \frac{1}{n} \sum_{k=0}^{n-1} \Delta_k^2 \leq \frac{2B_p^2 J(\mu_0)}{n},$$

which proves the stated rate. □

For the sampled data, Algorithm 1 minimizes

$$J^M(\boldsymbol{\omega}, c) := l^M(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} \phi_\gamma\big(w_p(\omega_n)|c_n|\big). \tag{49}$$

Its normalized insertion profile is

$$P^M_{p,\mu_t}(\omega) := \frac{P^M_{\mu_t}(\omega)}{w_p(\omega)}.$$

For fixed nodes, introduce the normalized coefficients $u_n := w_p(\omega_n)c_n$. The correction of the outer weights then minimizes

$$l^M\left(\sum_{n=1}^{N}\frac{u_n}{w_p(\omega_n)}\delta_{\omega_n}\right)+\alpha\sum_{n=1}^{N}\phi_\gamma(|u_n|).$$

The insertion step. The rate in Corollary 5.2 requires an exact maximizer of $|P^M_{p,\mu_t}(\omega)|$. Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2, Section 3.3]. The practical multistart search is proposed in the PDAP algorithm in the paper [21]. At the current iterate $\mu_t = \mu_{\boldsymbol{\omega}^{(t)},c^{(t)}}$ a candidate ω is therefore accepted when

$$\Delta(\mu_t,\omega)>0, \qquad \text{that is,} \qquad |P^M_{p,\mu_t}(\omega)|>\alpha L_\phi. \tag{50}$$

Theorem 5.1 provides two additional pieces of information. First, it gives the bound $R(\mu_t)$ for sampling the initial points at each iteration. Second, it gives an initial outer weight that guarantees a single-atom insertion decrease before the local semismooth Newton correction.

For a nonhomogeneous activation set $R_{\text{search}}(\mu_t) := \min\{R(\mu_t), e^5\}$. Each multistart initial point has norm at most $R_{\text{search}}(\mu_t)$. Starting from each ω_0, one unconstrained L-BFGS solve jointly optimizes all components of ω for $|P^M_{p,\mu_t}(\omega)|$. After optimization, candidates with $|\omega| > R_{\text{search}}(\mu_t)$ are discarded. This avoids constrained optimization.

Before running the SSN solver, the algorithm calculates

$$A^M_p(\omega) := \left\|\frac{K(\omega)}{w_p(\omega)}\right\|^2_M, \qquad \kappa_A(\omega) := -\frac{\Delta(\mu_t,\omega)}{A^M_p(\omega)}\operatorname{sign}\big(P^M_{p,\mu_t}(\omega)\big), \qquad c_A(\omega) := \frac{\kappa_A(\omega)}{w_p(\omega)}. \tag{51}$$

The coefficient $c_A(\omega)$ is used to initialize the subsequent semismooth Newton correction, which optimizes all outer weights. The decrease statement holds for every accepted candidate, not only for a global maximizer of $|P^M_{p,\mu_t}|$.

Writing $h_\gamma(z) := \phi_\gamma(|z|) - |z|$ and $\tau = (1+\alpha\gamma)^{-1}$, the semismooth Newton solve uses the coordinatewise residual

$$u = \operatorname{prox}_{\tau\alpha|\cdot|}\left(u-\tau\left[\nabla_u l^M\left(\sum_{n=1}^{N}\frac{u_n}{w_p(\omega_n)}\delta_{\omega_n}\right)+\alpha\nabla h_\gamma(u)\right]\right). \tag{52}$$

After the solve, the physical outer weights are recovered as $c_n = u_n/w_p(\omega_n)$; the proximal map in (52) is ordinary soft thresholding.

Algorithm 1 uses the scalar penalty ϕ_γ from (33) and the objective (49), with the insertion condition (50).

Algorithm 1 Iterative node insertion for J^M

Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
while $t < T_{\text{out}}$ **do**
  Set $R_{\text{search}} = \min\{R(\mu_t), e^5\}$. Sample N_{trial} points ω_0 with norm bounded by R_{search}.
  From each ω_0, run one unconstrained joint L-BFGS maximization of $|P^M_{p,\mu_t}(\omega)|$.
  Discard candidates with $|\omega| > R_{\text{search}}$, then remove near-duplicates.
  Keep the candidates satisfying (50); terminate if there is none.
  Insert at most N_{ins} nodes with the largest $\Delta(\mu_t,\omega)$, warm-start outer weight by (51).
  Optimize the outer weights of the enlarged network by the semismooth Newton method. If the correction increases J^M, discard it and keep the weights obtained immediately after insertion.
  Remove nodes with $|c_n| \le \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
  $t = t + 1$.
end while

5.2. **Algorithm 2: the fractional-exponent penalty on the sphere.** Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1)$, on the unit sphere. For $q < 1$, insertion condition of Algorithm 1 is unavailable (Remark 4.9). Since the zero network is a local minimizer, the bound on nonzero atom weights in Theorem 4.8 can guide initialization away from zero. We use another approach: for each candidate atom, we globally minimize the objective change over its coefficient and accept the insertion only if this change is negative. This global minimization finds a decreasing nonzero coefficient whenever one exists for the chosen atom, allowing a finite jump out of the local minimum at zero.

We use the scalar proximal map

$$\operatorname{prox}_{\lambda|\cdot|^q}(v) \in \arg\min_{z\in\mathbb{R}} \left\{ \frac{1}{2}(z-v)^2 + \lambda|z|^q \right\}, \tag{53}$$

with zero selected when the minimum is attained both at zero and at a nonzero point. For $0 < q < 1$, put

$$\bar{z} = [2\lambda(1-q)]^{1/(2-q)}, \qquad \bar{v} = \frac{2-q}{2(1-q)}\bar{z}.$$

Then the selected proximal value is zero when $|v| \le \bar{v}$; when $|v| > \bar{v}$, it is $\operatorname{sign}(v)z$, where z is the largest positive solution of

$$z + \lambda q z^{q-1} = |v|.$$

The implementation evaluates this value in closed form for $q \in \{1/2, 2/3\}$, using respectively a cubic in $\sqrt{z}$ and a quartic in $z^{1/3}$. At the endpoint $q = 1$, (53) is ordinary soft thresholding.

For a candidate $\omega \notin \operatorname{atom} \mu_t$, write

$$P := P^M_{\mu_t}(\omega), \qquad A := \|K(\omega)\|^2_M.$$

If A or $|P|$ is numerically zero, the candidate is discarded. Otherwise the exact one-atom objective increment is

$$\Delta J_{\psi_k}(c;\omega) = cP + \frac{1}{2}Ac^2 + \alpha|c|^q. \tag{54}$$

Completing the square in (54) gives its selected global minimizer directly:

$$c^*(\omega) := \operatorname{prox}_{(\alpha/A)|\cdot|^q}\left(-\frac{P}{A}\right). \tag{55}$$

We accept the candidate precisely when $\Delta J_{\psi_k}(c^*(\omega);\omega) < 0$. Thus the same coefficient both certifies the decrease produced by the insertion and initializes the subsequent correction.

After insertion, let m_t be the smallest magnitude among the outer weights. For $q < 1$, the correction fixes the proximal scale

$$\lambda_t := \min\left\{ \alpha, \frac{\rho_{\text{prox}} m_t^{2-q}}{2(1-q)} \right\}, \qquad \rho_{\text{prox}} = 0.5, \tag{56}$$

for the whole semismooth Newton solve. This choice ensures that every starting coefficient is a possible nonzero output of the proximal map, since $\bar{z} \le \rho_{\text{prox}}^{1/(2-q)} m_t < m_t$. Define, coordinatewise,

$$c(z) := \operatorname{prox}_{\lambda_t|\cdot|^q}(z), \qquad G_t(z) := \frac{\alpha}{\lambda_t}\big(z - c(z)\big) + \nabla_c l^M(\mu_{\boldsymbol{\omega},c(z)}). \tag{57}$$

The coefficient correction applies semismooth Newton to $G_t(z) = 0$. For $q = 1$ we use soft thresholding with $\lambda_t = \alpha$. For $q < 1$, an exact undamped Newton iteration on a fixed smooth branch has superlinear convergence rate locally under nonsingularity.

The implementation uses $k \in \{2, 3\}$, corresponding to $q \in \{2/3, 1/2\}$. The ReLU–ℓ^1 baseline reported alongside it is not this algorithm at $k = 1$. It is run separately and uses insertion when the profile magnitude exceeds α, initialization by coordinate descent, and the convex penalty $\alpha \sum_i |c_i|$.

Algorithm 2 Iterative node insertion for J_{ψ_k}

Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
while $t < T_{\text{out}}$ **do**
  Sample N_{trial} nodes randomly as the initial points.
  For each initial point, run unconstrained L-BFGS w.r.t. $|P|$ and normalize onto $\mathbb{S}^d$.
  Remove candidates that coincide numerically with the current support.
  For each candidate compute $c^*(\omega)$ from (55), and keep it when $\Delta J_{\psi_k}(c^*(\omega);\omega) < 0$; terminate if there is none.
  Retain at most N_{ins} candidates with the smallest $\Delta J_{\psi_k}(c^*;\omega)$, insert with outer weight c^*.
  Apply the semismooth Newton correction (57), using (56). If the correction increases J_{ψ_k}, discard it and keep the inserted coefficients c^*.
  Remove nodes with $|c_n| \le \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
  $t = t + 1$.
end while

6. Numerical examples

The experiments use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the stored sample rows, with 90% used for training and 10% for validation. Input and value normalization are computed from the full sample before the split.

For a fitted value function $\widehat{V}$, the relative errors are

$$\mathrm{Err}_{L^2} := \frac{\|\widehat{V} - V\|_{L^2}}{\|V\|_{L^2}},$$
$$\mathrm{Err}_{H^1} := \frac{\left(\|\widehat{V} - V\|_{L^2}^2 + \|\nabla_x \widehat{V} - \nabla_x V\|_{L^2}^2\right)^{1/2}}{\left(\|V\|_{L^2}^2 + \|\nabla_x V\|_{L^2}^2\right)^{1/2}}.$$

Both norms are computed empirically on the validation or evaluation set specified for each experiment with the original scale before normalization.

6.1. Van der Pol Oscillator.

We consider the optimal control of the Van der Pol oscillator:

$$\min_{u \in L^2(0,T;\mathbb{R})} \mathcal{C}(u;0,x) = \int_0^T \left(y_1^2(t) + y_2^2(t) + \eta u^2(t)\right) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta} g(y^*)^\top p^* = -\frac{1}{2\eta} p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$\begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - {y_1^*}^2) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^* y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - {y_1^*}^2), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases} \tag{58}$$

We solve (58) by adaptive collocation using SciPy's `solve_bvp`, with a residual tolerance of 10^{-8}. The resulting dataset contains 900 value–gradient samples on a 30×30 grid over $[-3, 3]^2$, split into 810 training and 90 validation samples.

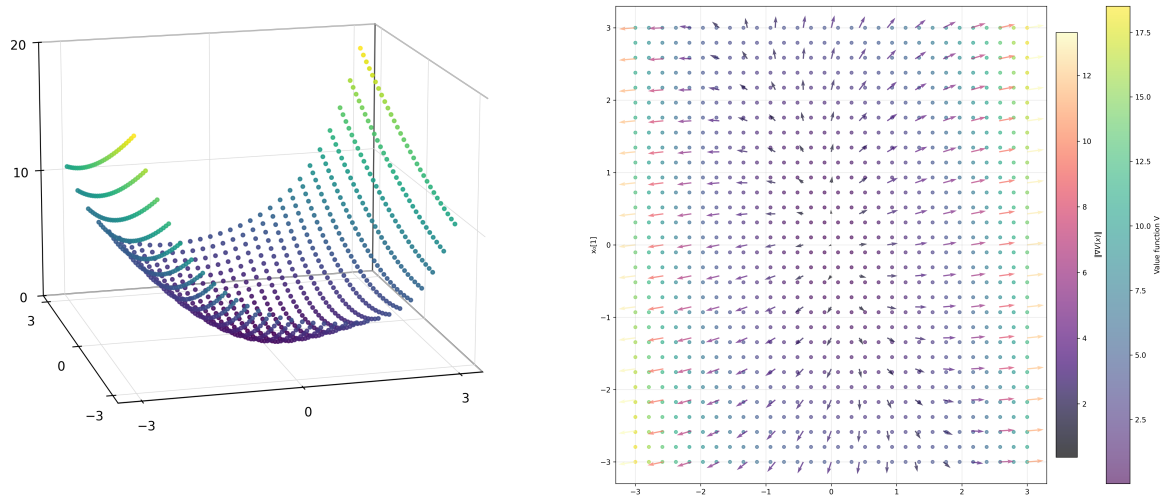


FIGURE 2. Van der Pol numerical value and gradient samples over $[-3, 3]^2$. Left: values v^m. Right: v^m (dot color) and costates g^m (arrows); arrow color indicates $|g^m|$.

Algorithms 1 and 2 use at most $T_{\text{out}} = 150$ outer iterations and $N_{\text{trial}} = 50$ sampled candidate nodes per iteration. Each iteration inserts the single selected candidate ($N_{\text{ins}} = 1$), so the decrease statement for insertion of one candidate applies. The loop terminates when no candidate clears the insertion threshold. The separate traditional ReLU–ℓ^1 comparators use 10 outer iterations with batch insertion of at most 15 candidates per iteration, from 50 candidate starts.

Algorithm 1: the log penalty on the normalized measure. The moment order enters the objective through the weight $w_p(\omega) = 1+|\omega|^p$ inside the scalar penalty: an atom at ω is charged $\phi_\gamma\big(w_p(\omega)|c|\big)$ rather than $\phi_\gamma(|c|)$, so a fixed coefficient becomes more expensive the further the atom sits from the origin. The exponent p is therefore an active parameter of the objective.

We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42, for softplus, tanh, the Gaussian and GELU^2, over

$$\alpha \in \{10^{-4}, 10^{-5}\}, \qquad p \in \{2.01, 2.5, 3, 4\}.$$

1. Effect of gradient training: We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

Switching from L^2 to H^1 training reduces the relative H^1 error from 0.223 to 0.0311 for softplus, from 0.241 to 0.0218 for tanh, and from 0.182 to 0.0111 for the Gaussian. Value accuracy improves as well (Table 1). The Gaussian reaches relative H^1 error 0.0111 with 32 atoms; tanh reaches 0.0218 with 29, and softplus 0.0311 with 14.

2. Choice of activation function: We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of $\nabla_x V$. Figure 3 contrasts the three activations through their value, first derivative, and second derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line,

TABLE 1. Effect of gradient-augmented fitting with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$.

Activation	L^2 trained	H^1 trained
softplus	8.38×10^{-2}	1.24×10^{-2}
tanh	7.59×10^{-2}	5.85×10^{-3}
Gaussian	5.28×10^{-2}	2.09×10^{-3}

(A) Effect of gradient training on Err_{L^2}

Activation	L^2 trained	H^1 trained
softplus	2.23×10^{-1}	3.11×10^{-2}
tanh	2.41×10^{-1}	2.18×10^{-2}
Gaussian	1.82×10^{-1}	1.11×10^{-2}

(B) Effect of gradient training on Err_{H^1}

and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\,\rho'(a \cdot x + b)$ in the gradient fitting problem.

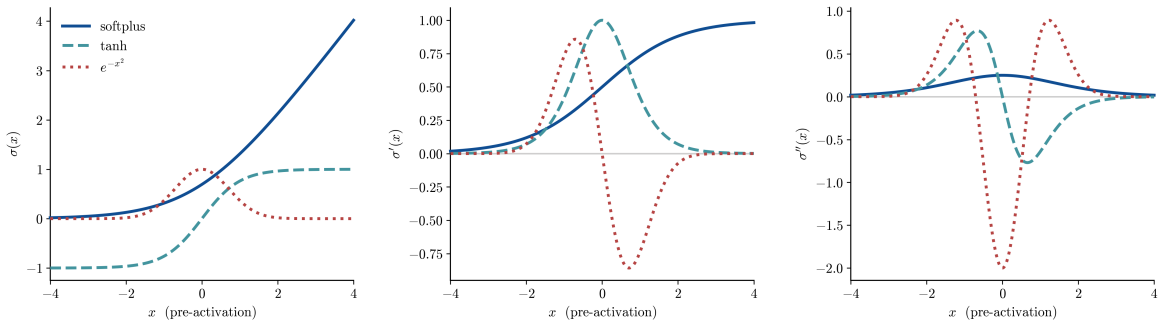
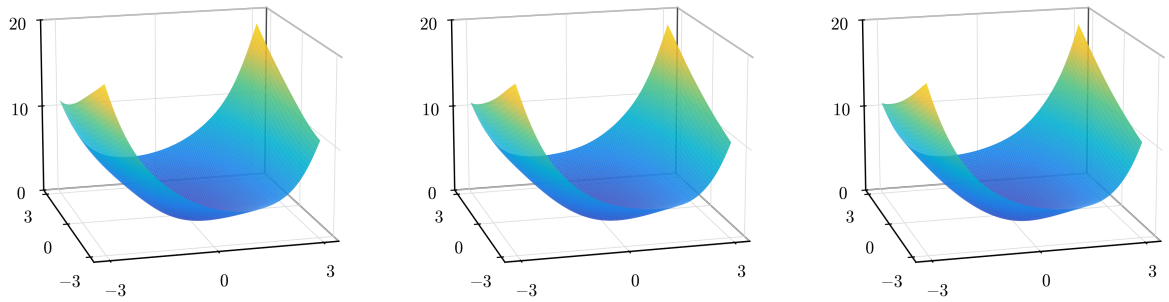


FIGURE 3. The three smooth activation functions used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-z^2} — shown as ρ, ρ', and ρ''.

For the tested parameters, the Gaussian has the smallest relative H^1 validation error, while softplus uses the fewest atoms. Thus accuracy and support size give different rankings.



(A) softplus ($N = 14$, $\mathrm{Err}_{H^1} = 0.0311$).

(B) tanh ($N = 29$, $\mathrm{Err}_{H^1} = 0.0218$).

(C) Gaussian ($N = 32$, $\mathrm{Err}_{H^1} = 0.0111$).

FIGURE 4. Learned value functions under the H^1 loss using the parameter values in Table 1. Predictions are clipped to $[0, 20]$ for visualization.

3. The computable search radius: The theoretical radius reduces the search region whenever $R(\mu_t) < e^5$, the fixed cap used in the implementation. In the experiments, the radius reduces the search region more often as p increases, from none of the eight runs at $p = 2.01$ to seven of eight at $p = 4$.

TABLE 2. Support size and validation error for the parameter values in Table 1.

Activation	N	Err_{L^2}
softplus	6	8.38×10^{-2}
tanh	10	7.59×10^{-2}
Gaussian	10	5.28×10^{-2}

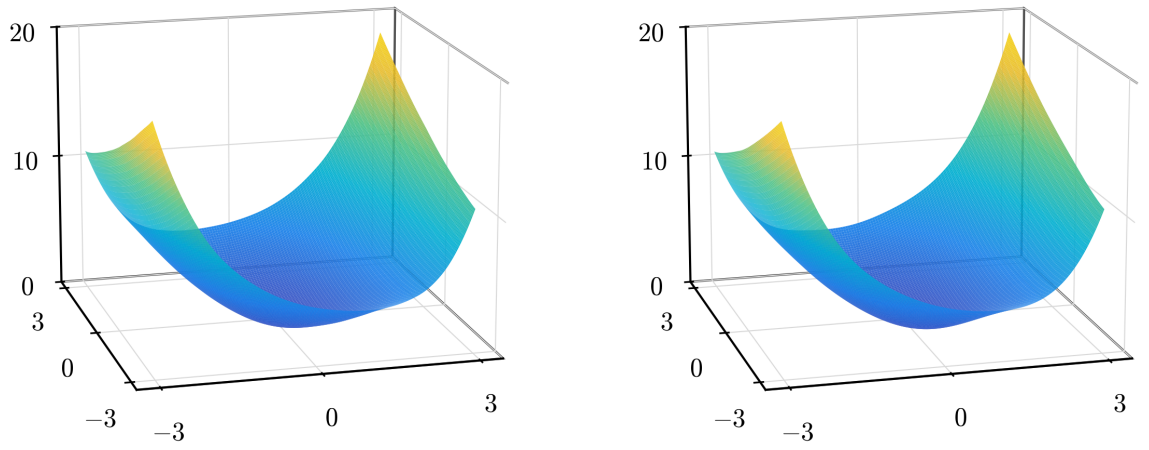
(A) L^2 trained

Activation	N	Err_{H^1}
softplus	14	3.11×10^{-2}
tanh	29	2.18×10^{-2}
Gaussian	32	1.11×10^{-2}

(B) H^1 trained

The recorded search radii fall below e^5 in thirteen of the thirty-two Van der Pol runs with different moment orders p: one at $p = 2.5$, five at $p = 3$, and seven at $p = 4$. The $p = 2.5$ case is the Gaussian at $\alpha = 10^{-4}$. At $p = 3$, both tanh and the Gaussian activate the smaller cap at both values of α, as does softplus at $\alpha = 10^{-4}$. At $p = 4$, all three activate it at both values of α, together with GELU^2 at $\alpha = 10^{-4}$.

Algorithm 2: $|c|^q$-penalty with ReLU^k activation. Algorithm 2 uses the activation function $\rho(z) = \mathrm{ReLU}(z)^k$ with the penalty ψ_k, namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k; all fits below use $\alpha = 10^{-5}$.



(A) $k = 2$
($N = 39$, $\mathrm{Err}_{H^1} = 0.0114$).

(B) $k = 3$
($N = 24$, $\mathrm{Err}_{H^1} = 0.0140$).

FIGURE 5. H^1-trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3\}$. Predictions are clipped to $[0, 20]$ for visualization; the reported errors use the unclipped network outputs.

For these fixed-α, adding gradient data lowers the relative H^1 error from 0.0813 to 0.0114 for $k = 2$, and from 0.0875 to 0.0140 for $k = 3$, factors of about 7.13 and 6.26, respectively (Table 3). Value error also decreases, while the support becomes larger.

Changing k changes both the activation and the induced exponent $q = 2/(k+1)$. At comparable H^1 error, the $k = 3$ fit uses 24 atoms instead of 39 for $k = 2$. Under the L^2 loss the fits use 14 and 13 atoms, respectively, with relative H^1 errors 0.0813 and 0.0875 (Table 3 (a)).

Figure 6 plots the best observed relative H^1 error at each support budget along the selected fits. At the shared Algorithm 1 operating point, softplus uses 14 atoms at error 0.0311 and the

TABLE 3. Algorithm 2: network size and validation error vs. the homogeneity degree k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	2.55×10^{-2}	8.13×10^{-2}
$k = 3$	13	2.79×10^{-2}	8.75×10^{-2}

(A) L^2 trained

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	39	2.75×10^{-3}	1.14×10^{-2}
$k = 3$	24	5.24×10^{-3}	1.40×10^{-2}

(B) H^1 trained

Gaussian uses 32 at 0.0111. At $\alpha = 10^{-6}$, the Algorithm 2 checkpoints selected for having the smallest recorded training objective in each run use 70 atoms at 0.00559 for ReLU^2 and 30 atoms at 0.00860 for ReLU^3.

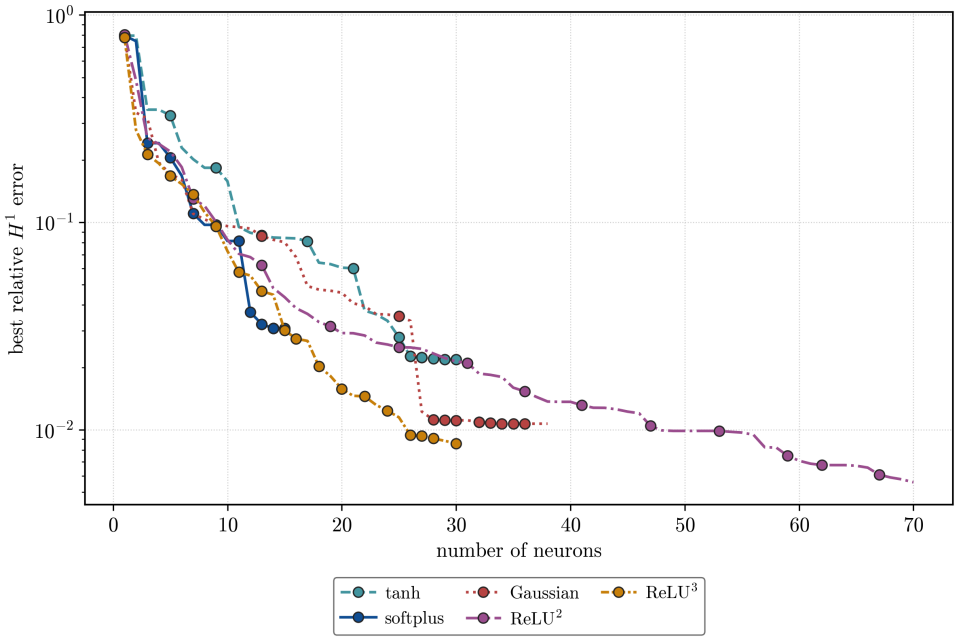
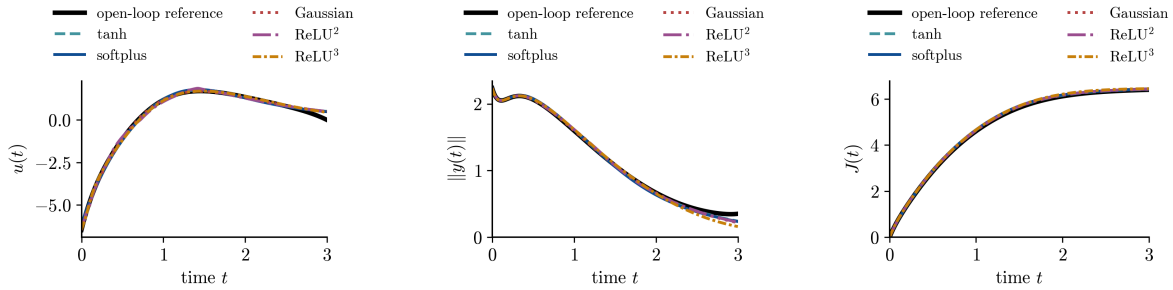


FIGURE 6. Best observed relative H^1 validation error against support budget. Nonhomogeneous activation functions use the parameter values stated in Table 1; homogeneous activation functions use $\alpha\psi_k$ at $\alpha = 10^{-6}$ for both $k = 2$ and $k = 3$.

Feedback law. We simulate trajectories under the explicit learned feedback laws from $y_0 = (2, 1)$ over $T = 3$, using fixed-step RK4 with $\Delta t = 0.01$, zero-order-hold controls, and the numerical bound $|u| \leq 50$. Figure 7 compares the signed control, state norm, and accumulated running cost for all five fitted controllers with a numerical open-loop reference. The learned laws have running costs between 6.415 and 6.449, compared with 6.401 for the numerical open-loop reference.

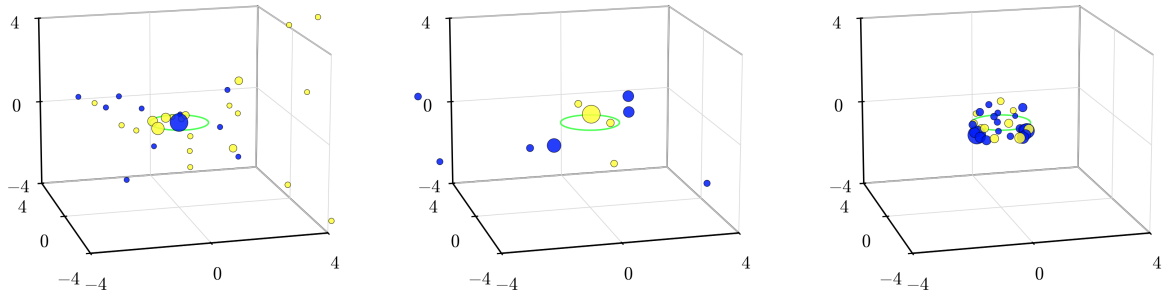
The compared networks have 29 neurons for tanh, 14 for softplus, 32 for the Gaussian, 70 for ReLU^2, and 30 for ReLU^3. The nonhomogeneous fits use $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$, and both homogeneous fits use $\alpha = 10^{-6}$.

Figure 8 shows the inner parameters for the Gaussian, softplus, and ReLU^k ($k = 3$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.



(A) Signed control $u(t)$. (B) State norm $\|y(t)\|$. (C) Accumulated cost $J(t)$.

FIGURE 7. Van der Pol controls, state norms, and realized costs from $y_0 = (2, 1)$ over $T = 3$. All three panels show the same five learned controllers and numerical open-loop reference (black). The accumulated cost $J(t) = \int_0^t [h(y(s)) + \eta|u(s)|^2]\, ds$ uses the RK4 stages for the learned laws, with the control held constant during each time step, and three-point Gauss quadrature on the dense PMP solution for the reference, without adding a learned terminal value.



(A) Gaussian. (B) softplus. (C) $\mathrm{ReLU}^3 + \alpha\,\psi_3$.

FIGURE 8. Inner parameters (a_1, a_2, b), with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$; dot color gives the sign and dot size the magnitude of the outer weight.

6.2. **Pendulum Swing-Up.** The question in this example is how a sparse network trained on values and gradients approximates a continuous value function whose gradient jumps across a switching set. We consider the pendulum example: we write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot\theta)^\top,$$

the pendulum angle and angular velocity. We consider the optimal control problem

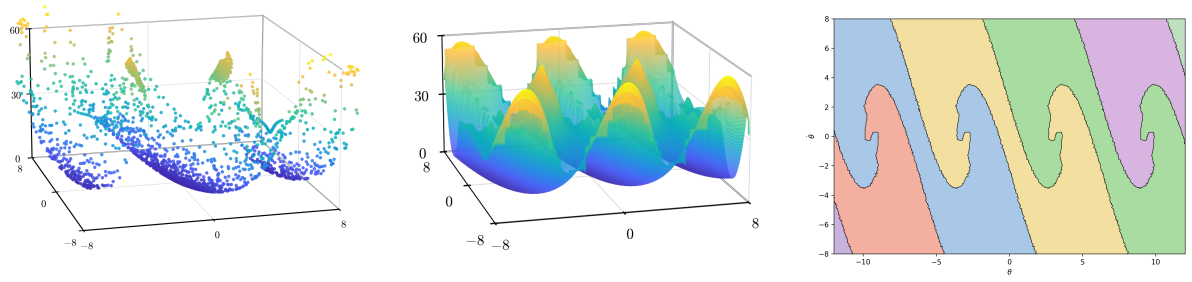
$$\min_{u \in L^2(0,\infty;\mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^\infty \left(\sin^2\theta(t) + (\cos\theta(t) - 1)^2 + \dot\theta^2(t) + \eta u^2(t)\right) dt$$

subject to

$$\begin{cases} m\ell^2\,\ddot\theta = -b\dot\theta + mg\ell\sin\theta + u, \\ (\theta(0), \dot\theta(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$. Here $m = 1$ is the pendulum mass, $\ell = 1$ the pole length, $b = 0.1$ the damping coefficient, $g = 9.8$ the gravitational acceleration, and $\eta = 1$; u is the applied torque.

We follow the open-loop algorithm of [11]. The state-value-gradient triple is generated for a value function with a discontinuous gradient.



(A) value samples $\{x^m, v^m\}$. (B) interpolated reference value. (C) Regions of attraction of the upright equilibria.

FIGURE 9. Numerical pendulum reference data over the $(\theta, \dot{\theta})$ plane. (a) Selected value samples from the PMP generator. (b) A periodic interpolation of those samples. (c) Regions of attraction of the upright equilibria.

Approximation. We use the same parameter grids as in the Van der Pol experiment. All networks are trained with the H^1 loss. We assess *numerical stabilization* using the following criterion: at every recorded sample in $[T-2, T]$, the trajectory must satisfy

$$\operatorname{dist}(\theta(t), 2\pi\mathbb{Z}) < 0.1, \qquad |\dot{\theta}(t)| < 0.5.$$

This finite-horizon criterion does not establish asymptotic stability. For model selection, we use $T = 40$. We select one representative network per model family for the plots below as follows:

(1) Across parameter settings, select the network with the smallest relative validation H^1 error set among those satisfying the stabilization criterion from both $A \approx (0.74, 0.61)$ and $B \approx (0.26, 0.46)$.
(2) If none satisfies the criterion, select the network with the smallest validation error and report its stabilization failure.

Figure 10 tracks the best observed relative H^1 validation error at each support budget. Among them, Gaussian and ReLU^2 provide the best balance between accuracy and sparsity. ReLU^2 achieves the highest accuracy, while Gaussian achieves comparable accuracy with fewer neurons. Both networks are smaller and more accurate than the selected tanh, ReLU^3, and softplus networks.

Figure 11 shows the fitted surfaces on the same physical window and value range as the value function in Figure 9b. The network shows limitation in approximating the optimal value function.

Feedback law. Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2\,\eta\, m\ell^2}\,\partial_{\dot{\theta}} V(x);$$

replacing V by a fitted $\widehat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.74,\ 0.61)$ and $B = (0.26,\ 0.46)$, integrating the dynamics up to $T = 40$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold (ZOH) controls, and the numerical bound $|u| \le 30$.

We chooose $T \in \{10, 20, 40\}$. From the harder start A, softplus, tanh, ReLU^2, and the ReLU–ℓ^1 baseline satisfy the numerical stabilization criterion at all three horizons; Table 4 reports $T = 10$ and $T = 40$ for both starts. From B, all five selected models pass at all three horizons; the

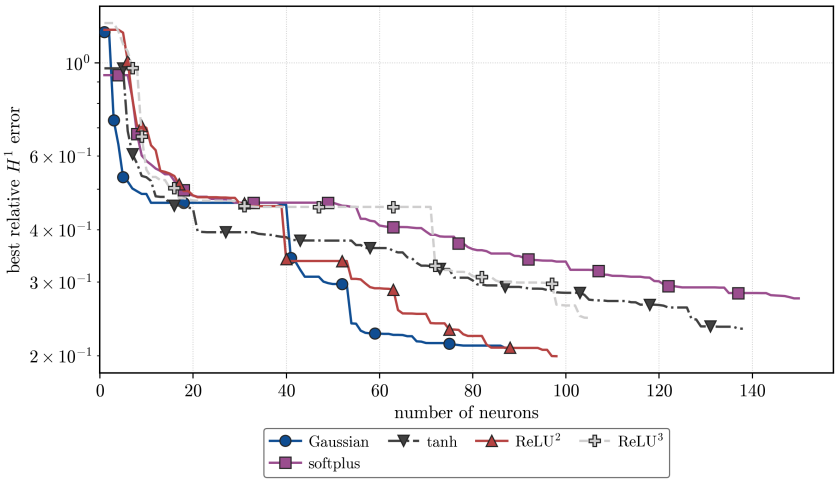
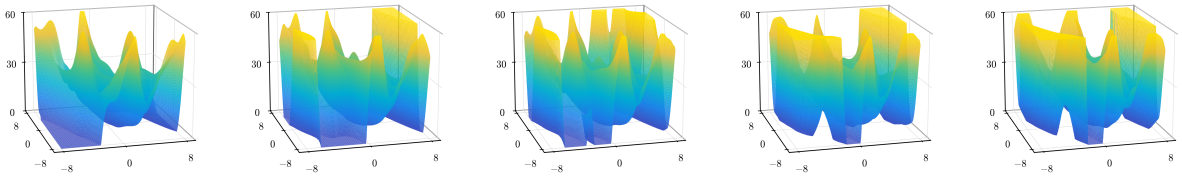


FIGURE 10. Best relative H^1 validation error against neuron count



(A) Gaussian (B) softplus (C) tanh (D) ReLU^2 (E) ReLU^3

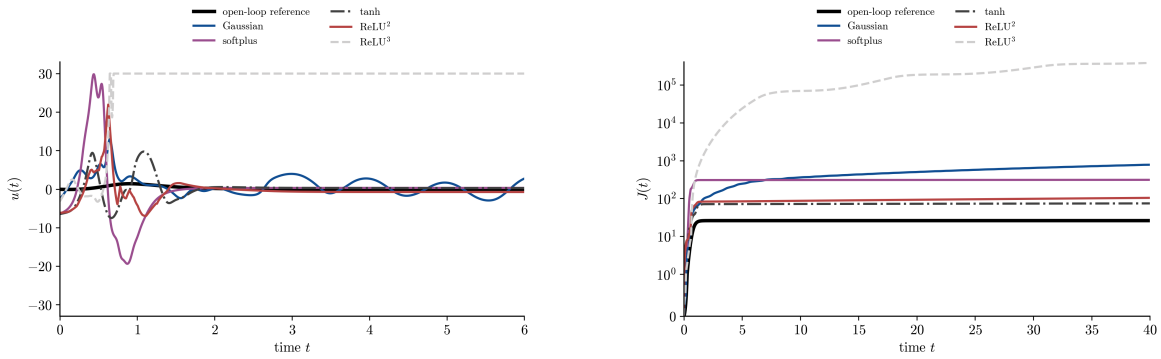
FIGURE 11. Learned value surfaces $\widehat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$. Predictions below 0 or above 60 are clipped to the respective boundary for visualization.

baseline passes at $T = 20, 40$ but not at $T = 10$. Halving the time step for integration and control updates preserves these classifications.

		Start A			Start B		
			Stabilized			Stabilized	
Feedback law	N	$J(10)$	$T = 10$	$T = 40$	$J(10)$	$T = 10$	$T = 40$
reference (PMP)	—	26.2	yes	yes	10.2	yes	yes
ReLU $+\ell^1$	72	82.2	yes	yes	115.7	no	yes
Gaussian	88	361.4	no	no	10.2	yes	yes
softplus	150	307.9	yes	yes	10.1	yes	yes
tanh	138	71.7	yes	yes	10.1	yes	yes
ReLU$^2 + \psi_2$	97	86.9	yes	yes	10.1	yes	yes
ReLU$^3 + \psi_3$	105	69306.3	no	no	10.1	yes	yes

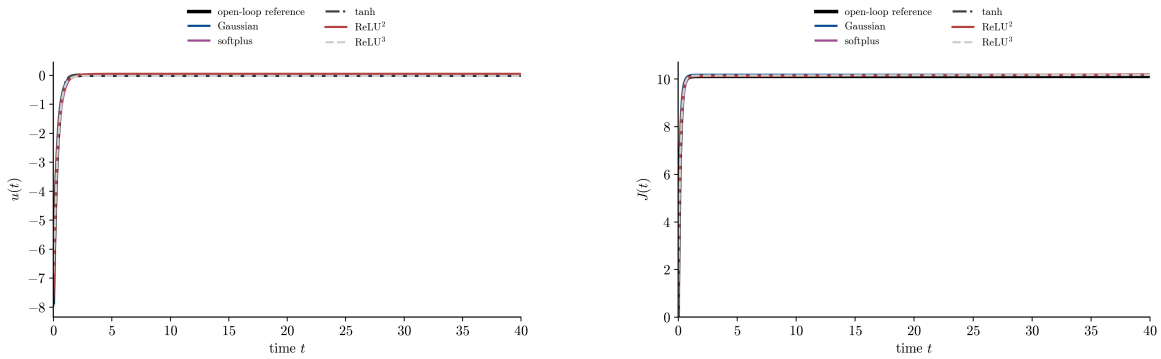
TABLE 4. Pendulum feedback from starts $A = (0.74, 0.61)$ and $B = (0.26, 0.46)$. N is the number of neurons.

Figures 12 and 13 compare all five learned controllers with numerical open-loop references from A and B with $T = 40$. Figure 12(a) focuses on $0 \le t \le 6$ to show the initial transient. The learned controls are clipped to $[-30, 30]$, and the flat segment of the ReLU^3 trace corresponds to saturation at the upper bound.



(A) Control for $0 \le t \le 6$. (B) Accumulated cost.

FIGURE 12. Pendulum controls and realized costs from $A = (0.74, 0.61)$.



(A) Control. (B) Accumulated cost.

FIGURE 13. Pendulum controls and realized costs from $B = (0.26, 0.46)$.

Validation errors near the gradient discontinuity. Let S denote the boundary across which the value function V remains continuous but its gradient jumps. To study approximation error near this gradient discontinuity, we consider the region within distance 0.3 of S:

$$S_{0.3} := \{x \in \mathbb{R}^2 : \mathrm{dist}(x, S) \le 0.3\},$$

where dist is Euclidean distance in the $(\theta, \dot{\theta})$ coordinates. Numerically, we use reconstructed points on S and copies shifted by $\pm 2\pi$ in the angle coordinate.

Figure 14 reports the relative H^1 error on a shared evaluation set of 994,558 points. Errors are reported on the subsets inside and outside $S_{0.3}$. The ReLU^2 model has the smallest errors in both regions (0.284 and 0.155). The corresponding values are 0.290 and 0.175 for the Gaussian, 0.331 and 0.194 for ReLU^3, 0.329 and 0.220 for tanh, and 0.363 and 0.250 for softplus. Every selected model has a larger relative error near the gradient discontinuity.

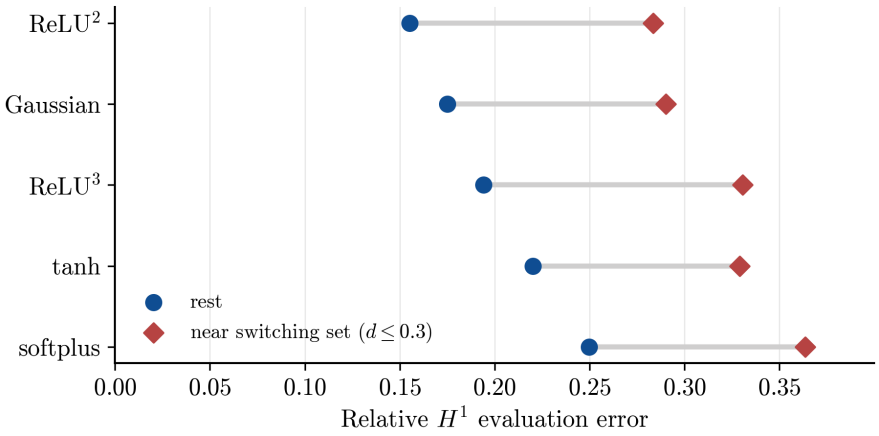


FIGURE 14. Relative H^1 error near the gradient discontinuity (within distance 0.3 of S, red diamonds) and on the remaining evaluation points (blue circles), with rows ordered by the latter error.

6.3. **Allen–Cahn equation.** Following Azmi, Kalise, and Kunisch [1, Section 3.3, (54)–(55)], we consider the optimal control of the Allen–Cahn equation on $\Omega = (-1, 1)$:

$$\min_{u \in L^2(0,T;\mathbb{R}^3)} \mathcal{C}(u; 0, y_0) = \int_0^T \left(\|y(t)\|^2_{L^2(\Omega)} + \beta \|u(t)\|_2^2 \right) dt$$

subject to

$$\begin{cases} \partial_t y - \nu \partial_x^2 y - y(1 - y^2) = \sum_{i=1}^{3} u_i(t) \mathbf{1}_{\omega_i}, & \text{in } (0, T) \times \Omega, \\ \partial_x y(t, -1) = \partial_x y(t, 1) = 0, & \text{in } (0, T), \\ y(0, x) = y_0, & \text{in } \Omega. \end{cases}$$

Here $\nu = 0.1$, $\beta = 0.01$, and the three controls act on $\omega_1 = (-0.7, -0.4)$, $\omega_2 = (-0.2, 0.2)$, and $\omega_3 = (0.4, 0.7)$; $\mathbf{1}_{\omega_i}$ denotes the indicator function.

For $T = 3$, we approximate this infinite-dimensional PDE control problem by a ten-dimensional ODE control problem using a cell-centered finite-volume spatial discretization [10, Section 4.1] with uniform cell width $\Delta x = 0.2$. Writing $y(t) \in \mathbb{R}^{10}$ for the vector of cell averages, the discretized problem has dynamics $\dot{y} = f(y) + Bu$ and running state cost $h(y) = \Delta x \, |y|^2$. Here f is the spatial discretization of $\nu \partial_x^2 y + y(1 - y^2)$ with zero boundary flux, and the columns of $B \in \mathbb{R}^{10 \times 3}$ contain the cell averages of $\mathbf{1}_{\omega_1}$, $\mathbf{1}_{\omega_2}$, and $\mathbf{1}_{\omega_3}$. We denote the discrete value function by $V_T(x)$, where $x \in \mathbb{R}^{10}$ is the initial state. Data generation and network fitting follow the same methods as the Van der Pol experiment in Section 6.1, with the same algorithmic iteration and candidate budgets. Here the data-generation residual tolerance is 10^{-6}, and the dataset consists of 2048 points in $[-1, 1]^{10}$, giving 1843 training and 205 validation samples.

Algorithm 1: the log penalty on the normalized measure. Gradient training improves both value and gradient approximation for all three smooth activations (Table 5). Switching from L^2 to H^1 training reduces the relative H^1 error from 0.128 to 0.0346 for softplus, from 0.154 to 0.0710 for tanh, and from 0.129 to 0.0563 for the Gaussian. Softplus shows the largest reduction. Value accuracy improves as well: its relative L^2 error decreases from 0.0452 to 0.0105, with improvements for the other two activations also visible in Table 5 (a).

Softplus combines the smallest network with the lowest H^1 error under gradient training (Table 6). Its 59 neurons give error 0.0346, whereas tanh uses 131 neurons at 0.0710 and the

TABLE 5. Effect of gradient-augmented fitting for the Allen–Cahn problem with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$.

Activation	L^2 trained	H^1 trained
softplus	4.52×10^{-2}	1.05×10^{-2}
tanh	5.18×10^{-2}	2.12×10^{-2}
Gaussian	4.18×10^{-2}	1.68×10^{-2}

(A) Effect of gradient training on Err_{L^2}

Activation	L^2 trained	H^1 trained
softplus	1.28×10^{-1}	3.46×10^{-2}
tanh	1.54×10^{-1}	7.10×10^{-2}
Gaussian	1.29×10^{-1}	5.63×10^{-2}

(B) Effect of gradient training on Err_{H^1}

Gaussian uses 148 at 0.0563. This changes the accuracy ranking observed in the Van der Pol experiment, where the Gaussian was the most accurate smooth activation and softplus was the sparsest (Table 2). Gradient training increases the support size for all three activations; for softplus, it grows from 29 to 59 neurons while improving both reported errors.

TABLE 6. Support size and validation error for the parameter values in Table 5.

Activation	N	Err_{L^2}
softplus	29	4.52×10^{-2}
tanh	59	5.18×10^{-2}
Gaussian	77	4.18×10^{-2}

(A) L^2 trained

Activation	N	Err_{H^1}
softplus	59	3.46×10^{-2}
tanh	131	7.10×10^{-2}
Gaussian	148	5.63×10^{-2}

(B) H^1 trained

Algorithm 2: $|c|^q$-penalty with ReLU^k activation. ReLU^2 is more accurate than ReLU^3 at nearly equal network sizes under both training losses (Table 7). Adding gradient data lowers the relative H^1 error from 0.107 to 0.0422 for $k = 2$ and from 0.171 to 0.0452 for $k = 3$, while also reducing the value errors. The corresponding support sizes grow from 59 to 143 and from 57 to 144 neurons. The higher homogeneity degree therefore gives no reduction in network size under H^1 training, whereas the $k = 3$ fit was smaller in the Van der Pol experiment (Table 3).

TABLE 7. Algorithm 2: network size and validation error vs. the homogeneity degree k for the Allen–Cahn problem.

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	59	3.01×10^{-2}	1.07×10^{-1}
$k = 3$	57	4.72×10^{-2}	1.71×10^{-1}

(A) L^2 trained

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	143	1.33×10^{-2}	4.22×10^{-2}
$k = 3$	144	1.43×10^{-2}	4.52×10^{-2}

(B) H^1 trained

Overall, softplus retains an accuracy advantage when the activations are compared at a common support budget (Figure 15). With at most 50 neurons, its best observed relative H^1 error is 0.0435, compared with 0.0749 for ReLU^2, 0.0838 for the Gaussian, 0.0850 for ReLU^3, and 0.106 for tanh. Thus the softplus error is about 42% smaller than that of the next-best activation at this budget. Its selected 59-neuron fit further reduces the error to 0.0346, combining a smaller support with better accuracy than the other selected fits.

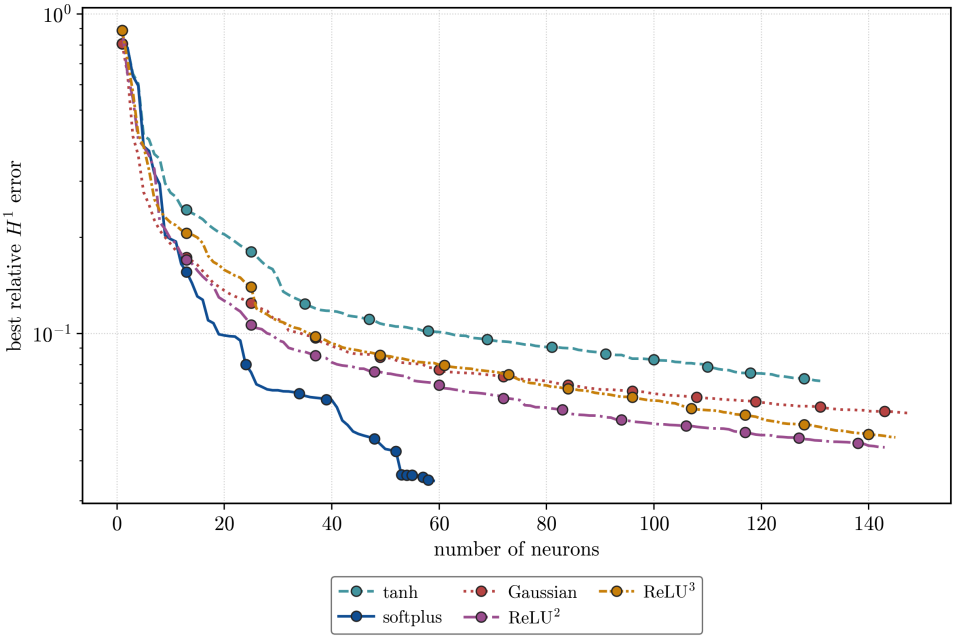
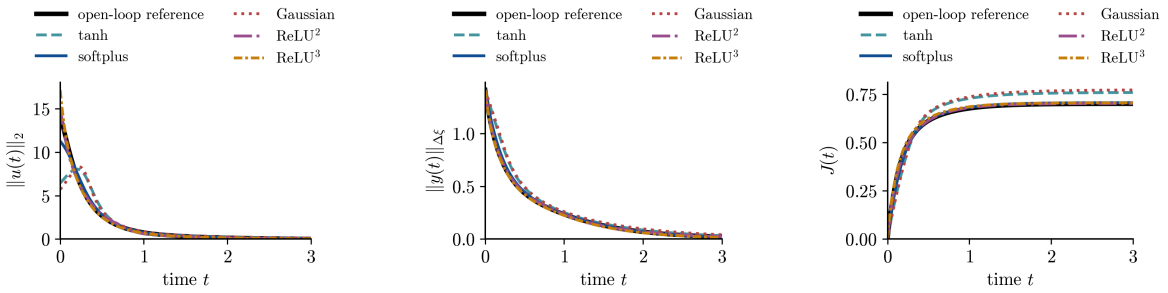


FIGURE 15. Best observed relative H^1 validation error against support budget for the Allen–Cahn problem. The construction and parameter choices follow Figure 6.

Feedback law. Among the learned controllers, ReLU^2 achieves the lowest realized cost from $y(0) = (1, \dots, 1)$ (Figure 16). We use the same feedback integration and running-cost evaluation as in Section 6.1 with

$$\widehat{u}(x) = -\frac{1}{2\beta} B^\top \nabla \widehat{V}_T(x), \qquad \|y\|_{\Delta x} = \sqrt{\Delta x\, |y|^2}.$$

All five learned laws reduce the spatial state norm from $\sqrt{2}$ to between 0.0144 and 0.0403 by $T = 3$. Their costs differ more clearly: ReLU^2 reaches 0.702, close to the PMP reference at 0.701, followed by ReLU^3 at 0.705 and softplus at 0.706. The corresponding costs for tanh and the Gaussian are 0.759 and 0.772.



(A) Control norm $\|u(t)\|_2$. (B) State norm $\|y(t)\|_{\Delta x}$. (C) Accumulated cost $J(t)$.

FIGURE 16. Control norms, spatial state norms, and realized costs from $y(0) = (1, \dots, 1)$ over $T = 3$. All three panels show the same five H^1-trained controllers as Figure 15 and the numerical open-loop PMP reference (black).

Comparison of six and ten state dimensions. Next we test the performance of the models in different dimensions. We fits the models with data in dimension 6 and 10, repectively. As Figure 17

shows, softplus remains the most accurate activation and exhibits the smallest error increase from six to ten state dimensions. The comparison keeps the sample budget fixed and selects the H^1-trained fit with the lowest validation error separately for each activation and dimension over the tested parameter grids. The softplus error rises from 0.0144 to 0.0226, compared with 0.0240 to 0.0422 for ReLU^2 and 0.0269 to 0.0452 for ReLU^3. The Gaussian and tanh errors more than double, from 0.0231 to 0.0548 and from 0.0277 to 0.0676, respectively. All five activations therefore lose accuracy at the higher dimension with this sample budget, and the Gaussian falls from second to fourth in the accuracy ranking, behind both homogeneous activations.

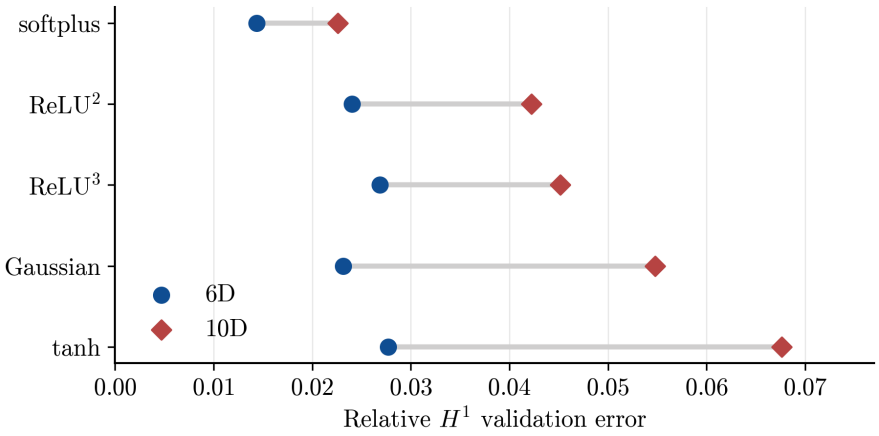


FIGURE 17. Relative H^1 validation errors of the best tuned H^1-trained Allen–Cahn networks in six (blue circles) and ten (red diamonds) state dimensions.

7. Conclusion and outlook

We developed a measure-theoretic framework for sparse, gradient-augmented recovery of HJB value functions from PMP data. Its main contribution is a variational route from infinitely wide dictionaries to finite shallow networks for two activation classes that require different compactness arguments. For nonhomogeneous activations, normalization of the measure and dictionary controls escape on the unbounded parameter domain and, together with the concave penalty, yields existence, finite support, separation and width bounds. For the single-atom PDAP scheme with exact search and monotone correction, we prove an $O(n^{-1/2})$ best-iterate bound on the first-order optimality residual for atom insertion. For positively k-homogeneous activations, radial optimization instead reduces the problem to the unit sphere and induces a fractional-exponent penalty. For ReLU^k with $k \geq 2$, the resulting formulation likewise admits global minimizers and explicit bounds on network width. These results give a rigorous foundation for constructing sparse networks that approximate both a value function and its gradient.

The experiments show that fitting both values and gradients improves the accuracy of sparse networks. For Van der Pol, the learned controllers achieve costs close to the numerical open-loop reference. For the pendulum, some of the learned controllers still fail to stabilize the system from the harder initial state. For Allen–Cahn, softplus gives the lowest approximation error and the smallest error increase from six to ten dimensions. The ReLU^2 controller also achieves a cost close to the reference for the tested initial state. These results show that sparse networks can provide accurate approximation of the value function and effective feedback, including in higher dimensions. The pendulum example also highlights their limitations near discontinuities in the value gradient.

The most immediate extension concerns the insertion step: the rate analysis assumes exact global candidate maximization, whereas the implementation uses a multistart local search. Developing certified or more reliable candidate searches would therefore bring the algorithm closer to the theoretical guarantee. A second direction is to complement the present approximation criteria with training or validation that accounts for feedback performance near switching sets.

References

[1] Behzad Azmi, Dante Kalise, and Karl Kunisch. Optimal feedback law recovery by gradient-augmented sparse polynomial regression. *Journal of Machine Learning Research*, 22(48):1–32, 2021.

[2] Francis Bach. Breaking the curse of dimensionality with convex neural networks. *Journal of Machine Learning Research*, 18(19):1–53, 2017.

[3] Francesca Bartolucci, Marcello Carioni, José A. Iglesias, Yury Korolev, Emanuele Naldi, and Stefano Vigogna. A Lipschitz spaces view of infinitely wide shallow neural networks. *SIAM Journal on Mathematical Analysis*, 58(3):2786–2828, 2026.

[4] Yoshua Bengio, Nicolas Le Roux, Pascal Vincent, Olivier Delalleau, and Patrice Marcotte. Convex neural networks. In *Advances in Neural Information Processing Systems*, volume 18, 2005.

[5] Guy Bouchitté and Giuseppe Buttazzo. New lower semicontinuity results for nonconvex functionals defined on measures. *Nonlinear Analysis: Theory, Methods & Applications*, 15(7):679–692, 1990.

[6] Piermarco Cannarsa and Halina Frankowska. Some characterizations of optimal trajectories in control theory. *SIAM Journal on Control and optimization*, 29(6):1322–1347, 1991.

[7] Frank H Clarke and Richard B Vinter. The relationship between the maximum principle and dynamic programming. *SIAM Journal on Control and Optimization*, 25(5):1291–1311, 1987.

[8] Emiliano Cristiani and Pierre Martinon. Initialization of the shooting method via the hamilton-jacobi-bellman approach. *Journal of Optimization Theory and Applications*, 146:321–346, 2010.

[9] Kenji Doya. Reinforcement learning in continuous time and space. *Neural Computation*, 12(1):219–245, 2000.

[10] Thierry Gallouët, Raphaèle Herbin, and Marie Hélène Vignal. Error estimates on the approximate finite volume solution of convection diffusion equations with general boundary conditions. *SIAM Journal on Numerical Analysis*, 37(6):1935–1972, 2000.

[11] Haoyu Han and Heng Yang. On the nonsmooth geometry and neural approximation of the optimal value function of infinite-horizon pendulum swing-up. In *Proceedings of the 6th Annual Conference on Learning for Dynamics and Control (L4DC)*, volume 242 of *Proceedings of Machine Learning Research*, pages 654–666. PMLR, 2024.

[12] Tjeerd Jan Heeringa, Len Spek, Felix Schwenninger, and Christoph Brune. Embeddings between barron spaces with higher order activation functions. *arXiv preprint arXiv:2305.15839*, 2024.

[13] Michael Hinze, René Pinnau, Michael Ulbrich, and Stefan Ulbrich. *Optimization with PDE Constraints*, volume 23 of *Mathematical Modelling: Theory and Applications*. Springer, Dordrecht, 2009.

[14] Wei Kang, Qi Gong, Tenavi Nakamura-Zimmerer, and Fariba Fahroo. Algorithms of data generation for deep learning and feedback design: A survey. *Physica D: Nonlinear Phenomena*, 425:132955, 2021.

[15] Yuanyuan Li, Shuai Lu, Peter Mathé, and Sergei V. Pereverzev. Two-layer networks with the ReLU^k activation function: Barron spaces and derivative approximation. *Numerische Mathematik*, 156:319–344, 2024.

[16] Derong Liu, Ding Wang, Fei-Yue Wang, Hongliang Li, and Xiong Yang. Neural-network-based online HJB solution for optimal robust guaranteed cost control of continuous-time uncertain nonlinear systems. *IEEE Transactions on Cybernetics*, 44(12):2834–2847, 2014.

[17] Michael Lutter, Boris Belousov, Kim Listmann, Debora Clever, and Jan Peters. HJB optimal feedback control with deep differential value functions and action constraints. In *Proceedings of the Conference on Robot Learning (CoRL)*, volume 100 of *Proceedings of Machine Learning Research*, pages 640–650. PMLR, 2020.

[18] Rémi Munos, Leemon C. Baird, and Andrew W. Moore. Gradient descent approaches to neural-net-based solutions of the Hamilton-Jacobi-Bellman equation. In *IJCNN'99. International Joint Conference on Neural Networks*, volume 3, pages 2152–2157. IEEE, 1999.

[19] Tenavi Nakamura-Zimmerer, Qi Gong, and Wei Kang. Adaptive deep learning for high-dimensional hamilton-jacobi-bellman equations. *SIAM Journal on Scientific Computing*, 43(2):A1221–A1247, 2021.

[20] Behnam Neyshabur, Ryota Tomioka, and Nathan Srebro. In search of the real inductive bias: On the role of implicit regularization in deep learning. *arXiv preprint arXiv:1412.6614*, 2014.

[21] Konstantin Pieper and Armenak Petrosyan. Nonconvex regularization for sparse neural networks. *Applied and Computational Harmonic Analysis*, 61:25–56, 2022.

[22] Konstantin Pieper and Daniel Walter. Linear convergence of accelerated conditional gradient algorithms in spaces of measures, 2021. arXiv:1904.09218v2 [math.OC].

[23] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[24] Alena Shilova, Thomas Delliaux, Philippe Preux, and Bruno Raffin. Revisiting continuous-time reinforcement learning. a study of HJB solvers based on PINNs and FEMs. In *16th European Workshop on Reinforcement Learning (EWRL)*, 2023.

[25] Yuval Tassa and Tom Erez. Least squares solutions of the HJB equation with neural network value-function approximators. *IEEE Transactions on Neural Networks*, 18(4):1031–1041, 2007.